

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf\_684a0618-ccb\agent-a94e5a7a964c80c58.jsonl`
- SHA-256: `f55ade4905597ac7da107f032bcae1078537fdcd9977be9e333bc738899a1c69`
- Source modified: `2026-06-19T19:19:33+00:00`
- Imported at: `2026-07-08T16:00:03+00:00`
- project: `wf\_684a0618-ccb`
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`

Transcript

1. user (2026-06-19T19:17:29.045Z)

THE IDEA TO ASSESS: a desktop app OR web UI where the user points KhelSutra at an ONLINE storage location they already own/pay for ? a Google Cloud Storage bucket, Google Drive, OneDrive/Microsoft 365, Dropbox, S3, etc. KhelSutra then WATCHES that folder, pulls/streams new recordings as they arrive, and AUTO-prepares highlights (same processing as the local/desktop path). "Bring your own storage + autosync": footage lands in their cloud ? reels appear automatically, zero manual upload.

You are assessing how this idea fits ONE area of the KhelSutra (badminton-highlight-indexer) docs. Read: docs/COMMERCIALIZATION.md, docs/PMF_MARKET_RESEARCH.md, docs/monetization (and docs/archives/MONETIZATION_AUDIT.md if present). Business fit of BYO-storage: it shifts storage cost + egress to the user and meets footage where it already auto-uploads (GoPro Quik ? cloud, phone ? Drive/Photos, academies on MS365). Who is the segment? Who pays egress if WE pull vs if the DESKTOP app pulls? Does BYO-storage strengthen the cost story (we don't store giant videos) or complicate licensing/privacy (we briefly hold their video)?

Be evidence-based and honest ? quote real anchors, don't invent. If the doc does NOT anticipate the idea, say so plainly and note what it WOULD take. Return the structured object.

2. assistant (2026-06-19T19:17:32.199Z)

I'll assess how the BYO-storage + autosync idea fits the KhelSutra business/strategy docs. Let me start by pulling the repo and reading the relevant docs.

3. user (2026-06-19T19:17:36.000Z)

Your configuration specifies to merge with the ref 'refs/heads/docs/court-floor-rally-end' from the remote, but no such ref was fetched.

4. user (2026-06-19T19:17:36.350Z)

ABLATION_LAYER.md
ANALYZERS_AND_RECONCILERS.md
BACKLOG.md
CODE_AUDIT_AND_TEST_HARDENING.md
CODE_MAP.md
COMMERCIALIZATION.md
DEFERRED_HARDENING_PLAN.md
DESKTOP_APP_PLAN.md
EVALUATION.md
EXECUTION_POLICY.md
EXPERIMENT_HARNESS.md
FIELD_VISIT_ANSWER_SHEET.md
FIELD_VISIT_PLAYBOOK.md
GOLDEN_DATA_SHARING.md

GOLDEN_REGRESSION_FIXTURES.md
GOLDEN_SET_IMPLEMENTATION.md
GOLDEN_SET_PHASE1_VIDEOS.md
GOLDEN_SET_VENDORING.md
GOLDEN_VIDEOS.md
HARDENING_LOOP_HANDOFF.md
HOSTING_PLAN.md
HOW_RALLY_DETECTION_WORKS.md
I18N_PLAN.md
KHELSUTRA_PRODUCT_OVERVIEW.md
MULTI_SIGNAL_FUSION_PLAN.md
NEXT_STEPS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PERSONALIZATION_PLAN.md
PER_VIDEO_OUTPUT_LAYOUT.md
PLATFORM_ARCHITECTURE.md
PMF_FIELD_SIGNALS.md
PMF_MARKET_RESEARCH.md
POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
PRODUCT_FIELD_ASSOCIATE_FLYER.md
PROJECT_DOC_TEMPLATE.md
PROMINENT_COURT_DETECTION.md
QUALITY_ITERATIONS.md
R1_MILESTONE_LAUNCH_REPORT.md
R2_EVAL_METRIC_DESIGN.md
RALLY_DETECTION_GAP_CLOSURE.md
RALLY_DETECTION_QUALITY_REPORT.md
RALLY_QUALITY_RESEARCH.md
README.md
REAL_FOOTAGE_VALIDATION_REPORT.md
ROADMAP.md
ROORA_LABELING_GUIDELINES.md
STORAGE_SHARING_MODEL.md
TRACKNET_WSL_SETUP.md
UI_ALPHA_EXECUTION_PLAN.md
UI_PROPOSAL.md
VIDEO_LOCALITY_MODEL.md
VIDEO_RECORDING_GUIDELINES.md
archives
mockups
---ARCHIVES---
CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
COMMERCIAL_READINESS_REVIEW.md
DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
MONETIZATION_AUDIT.md
README.md
TEST_RUN_SUMMARY.md
checkpoints
decisions
past_projects
quality-iterations
research

5. assistant (2026-06-19T19:17:41.528Z)

The pull failed because the branch is a local docs branch, but that's fine ? I'm reading docs that already exist locally. I notice there are additional relevant docs beyond what was named: `docs/STORAGE\_SHARING\_MODEL.md`, `docs/VIDEO\_LOCALITY\_MODEL.md`, `docs/HOSTING\_PLAN.md`, `docs/PLATFORM\_ARCHITECTURE.md`, `docs/DESKTOP\_APP\_PLAN.md`. Let me read the named docs plus these clearly-relevant ones.

6. user (2026-06-19T19:17:42.006Z)

1 # Commercialization Tracker

```

2
3  **Status:** ? LIVING ? track this in parallel with product/quality work.
4  **Owner:** avidullu . **Last refreshed:** 2026-06-07
5
6  **Purpose.** Single source of truth for the **commercial** dimension of the project:
7  licensing/legal posture, codec & output economics, hosting unit-economics, the
8  golden-set vendoring spend, packaging/distribution, and the product-market-fit
9  risks that gate monetization. Updated as decisions land ? see the
10 [Changelog](#changelog) at the bottom.
11
12  **Merged from / supersedes:**
13  - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14    audit) ? kept as the **research-of-record** (per-claim verification transcript,
15    primary sources). This doc carries its *current* status forward.
16  - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
17    archived; its commercial findings + the quality-phase priorities live here now.
18
19  **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20  users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21  distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22  **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23  royalties** all still apply.
24
25  ---
26
27  ## 1. Commercialization dashboard
28
29  Legend: ? done . ? in progress / watch . ? blocker / needs action . ?? needs legal
30  counsel . ? future
31  Priority: **P0** (do first) . **P1** . **P2** (lowest). Rows are grouped **Pending ? In-
32  flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
33  order.)
34
35  | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
36  |---|---|---|---|---|---|
37  | | **? PENDING (needs action / not started)** | | | | |
38  | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
39  AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
40  remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
41  HW encode. See §4. Decision: provision Ada-class GPU? |
42  | C9 | **Packaging / distribution** | **P0** | ? | `pyproject.toml` (PEP 621, hatchling)
43  added; the diagnostic `setup.py` renamed to `scripts/doctor.py`; `indexer` console entry point;
44  CI installs `pip install -e .[dev]`. torch/torchvision stay an out-of-band `--index-url` install
45  (PEP 621 can't pin CUDA wheels). | Implemented per `DEFERRED_HARDENING_PLAN.md` (#7). Remaining:
46  publish `obsreport` so the `reporting` extra is pip-installable. |
47  | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Async job model shipped
48  (`DEFERRED_HARDENING_PLAN.md` #16, PR #87); auth/tenancy still future. | Multi-user/platform
49  strategy in `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration). |
50  | | **? IN-FLIGHT (in progress / watch)** | | | | |
51  | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini
52  2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned
53  open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
54  "provider registry + paid fallback" a core principle; build the owned model **after** the
55  scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
56  | C13 | **Beachhead validation** | **P0** | ? | Market research
57  (`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead.
58  Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R
59  benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See
60  §6. |
61  | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)**
62  selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). |
63  Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
64  `GOLDEN_SET_VENDORING.md`. |
65  | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ?? | WASB-SBDT **code is MIT
66  (clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code

```

grant. MonoTrack/YOLO-NAS banned (noncommercial). | ****Ship-gating**** blocker (multiplies per sport). Resolve via an own-trained detector ****or**** a clean MIT/Apache detector swap ? NOTE: ADR-002 says our temporal labels can't train the detector (needs coordinate labels), so own-weights is a new, un-scoped workstream, NOT delivered by ROADMAP Phase 4. |

43 | C14 | ****Owned-model base licensing + training-data policy**** | ****P1**** | ? ?? | Owned video-QA model must build on a ****commercial-clean (MIT/Apache-2.0/BSD), video-capable**** base ? primary ****Qwen3-VL (Apache-2.0)****, fallback ****InternVL3 (MIT)****. ****Gemma 4 = AMBER / bench-only**** (Apache grant likely but Prohibited-Use posture unconfirmed ? counsel needed; not a build target yet). ****Reject Gemma 3**** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), ****gpt-oss**** (text-only). ****Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs**** (competing-model clause). The moat is the ****consented dataset + eval harness****, not the model. | ****MIT/Apache-2.0/BSD**** for code+data+weights; ****open-model teachers only****; separate explicit ****training opt-in****; ****exclude minors****; per-clip consent ledger. See ****OWNED_MODEL_IMPLEMENTATION_PLAN.md**** (authoritative). |

44 | C6 | ****Inbound HEVC decode**** | ****P2**** | ? ?? | Decoding users' GoPro H.264/****HEVC**** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |

45 | C7 | ****Hosting unit-economics**** | ****P2**** | ? | Per-video cost is ****cheap**** for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | ****Benchmark \$/video & latency on the target VM**** for A/B/C before pricing. See §5. |

46 | C10 | ****Value layer (product surface)**** | ****P2**** | ? | Minimal exports shipped (EDL/JSON/CSV + ****basic_stats****, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

47 | C11 | ****Input-quality enforcement**** | ****P2**** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more ****VIDEO_RECORDING_GUIDELINES**** dimensions into validators/report rules. |

48 | C2 | ****`supervision` ByteTrack pin**** | ****P2**** | ? | Pinned ****`<0.30.0`** (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |

49 | | ****COMPLETED**** | | | |

50 | C1 | ****AGPL / copyleft removal**** | ? | ? | ADR-005 shipped: AGPL ****`ultralitics` YOLOv8`** replaced with **\*\*torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)****. No AGPL code/weights in the stack. | Done. Keep new deps permissive. |

51

52 ---

53

54 ## 2. Licensing & dependency posture (refreshed risk table)

55

56 Verdicts updated against the current stack (****`requirements.txt` + code****). Original

57 2026-05-30 verdicts and the 25-claim verification transcript are in

58 ****`archives/MONETIZATION\_AUDIT.md`**.

59

60 | Dependency | License | SaaS obligation? | Current verdict | Notes |

61 |---|---|---|---|---|

62 | ****`~`ultralitics` + `yolov8n.pt`~`** | AGPL-3.0 | Yes (§13) | ? ****REMOVED**** | Replaced per ADR-005 ? no longer in the stack. |

63 | ****torchvision**** detectors (Faster R-CNN/RetinaNet) | BSD-3 | No | ? LIVE | The chosen replacement; person = COCO class (see ****`backend/pipeline/detectors/person\_detector.py` `\_PERSON\_CLASS\_ID`**). |

64 | ****`supervision`**** (ByteTrack) | MIT | No | ? LIVE | Pinned ****`<0.30.0`** ? migrate before bump (C2). |

65 | ****WASB-SBDT**** (shuttle) | MIT (code) | No | ? code clear | Weights provenance open (C3). |

66 | ****MonoTrack / YOLO-NAS**** | noncommercial | ? | ? BANNED | Do not use. |

67 | ****Gemini API**** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH | Must bill the paid tier to keep user video private (C4). |

68 | H.264 / HEVC / AAC codecs | patent pools | royalties on encode/decode | ? ?? | Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6). |

69 | ffmpeg/ffprobe, google-genai, torch, opencv, numpy, fastapi/uvicorn/pydantic/jinja2/dotenv | LGPL/Apache/BSD/MIT | No | ? CLEAR | Invoked as subprocess / imported; all permissive. |

70 | ****`obsreport`**** (reports) | permissive (private) | No | ? soft dep | Pin the ****`git+ssh@vX`** tag once the repo is published (BACKLOG). |

71

72 ****Bottom line:**** the two hard blockers from the audit (Ultralytics; banned weights)

```

73 are resolved/avoided. Remaining commercial-legal items are **codec royalties (C5/C6)**
74 and **weights provenance (C3)**, plus the metered-API privacy posture (C4).
75
76 ---
77
78 ## 3. Runtime AI strategy ? A/B/C
79
80 The semantic rally-boundary step is the one external/metered dependency. Three paths
81 (from the audit), with current standing:
82
83 - **(A) Eliminate ? offline learned segmenter.** ActionFormer (MIT) / our own learned
84 classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call
85 cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB
86 shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
87 - **(B) Self-host VLM.** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No
88 per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
89 - **(C) Keep Gemini 2.5 Flash (paid).** Fastest, cheap per video, video-native;
90 ongoing metered cost + external dependency. **? current live path.**
91
92 Hosted alternatives for (C) comparison: Anthropic Claude (no training on
93 inputs/outputs; customer owns outputs; **no native video** ? weaker here);
94 OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
95
96 ---
97
98 ## 4. Codec / output economics (the live gap ? C5)
99
100 **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
101 codecs it implements. Royalties attach to the act of encode/decode.
102
103 - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
104 is first 100k units/yr royalty-free with a permanent "free to end-users over the
105 internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
106 - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
107 (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
108 must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110 highest uncertainty; low residual.
111
112 **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113 output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114 *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116 ---
117
118 ## 5. Hosting unit-economics (C7)
119
120 Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122 - **(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123 tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
124 economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
125 - **(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
126 VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
127 - **(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
128 + light GPU. **Best steady-state margin.**
129
130 > The A/B/C $/video comparison depends on GPU instance + batch throughput ?
131 > **benchmark on the target VM before pricing the product.**
132
133 ---
134
135 ## 6. Product-market fit & the quality/data phase
136
137 ### Beachhead (from market research, 2026-06-07 ? see

```

```
[`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
138     The static-camera constraint is a B2B filter: the buyer must already record from a
139     fixed angle, which points away from the median casual player and toward coaches/clubs.
140     Ranked beachheads:
141     1. Badminton coaching academies & serious players, India (strongest) ? 20M+
players,
142     formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
143     who already monetize, badminton whitespace vs incumbents. Sell to the academy,
not
144     the individual; needs INR-tuned pricing and a Kannada/Hindi product ? see
145     [I18N_PLAN.md](I18N_PLAN.md) (Bangalore-first localization, by design).
146     2. US/global pickleball players & clubs ? fastest-growing sport (19.8M US, +45.8%
YoY)
147     but SwingVision already owns single-camera tennis/pickleball AI (USD 179.99/yr) ?
148     head-to-head, not greenfield.
149     3. Badminton academies/clubs in East/SE Asia + Denmark ? highest density; China
needs
150     local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.
151
152     Pricing anchors (estimate): software-only BYO-camera B2C USD 60?180/yr, B2B
153     academy/club USD 300?1,500/yr per location (undercuts hardware-bundled Veo/Pixellot/
154     Spiideo; comparable to SwingVision software-only).
155
156     PMF risks (from the readiness review)
157     - Capture quality is the gate. Accuracy is capture-conditional; broadcast/handheld
158     footage is a failure mode. The wedge (static, tripod, full-court) is defensible but
159     narrows the addressable market ? enforce/guide it (C11, VIDEO_RECORDING_GUIDELINES).
160     - Value beyond raw segments is table stakes. Time-saved editing alone is weak;
161     per-rally tags, stats, and a correction loop are what make it sticky (C10).
162     - Golden labels are the recurring blocker across the offline segmenter, audio
163     fusion, and measurable quality ? the highest-leverage investment (C8).
164
165     Refined priority order (quality/data phase ? "focus on quality once new videos
land")
166     Carried from the readiness review; commercialization items above are tracked in
parallel.
167
168     1. Golden-set expansion + data-flywheel UX (highest with new labeled videos):
169     ingest the new batch, expand calibration/CV/regression baselines, make label
170     contribution + safe retraining easy and non-contaminating (eval/training separation).
171     2. Hybrid unification + observability hooks (deferred item 7, now higher): extract
172     shared logic from tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics
land.
173     3. Detector/windowing quality iteration with new data: re-tune velocity/merge/
174     min-duration + rally_gate`; validate WASB on real video; trial audio fusion
(ADR-007).
175     4. Deeper value layer (beyond minimal exports, C10): per-rally tagging/notes ?
176     candidate_segments`, per-player/match stats, clip gallery, human-correction loop.
177     5. Input-quality enforcement & guidance polish (C11): more guideline dimensions ?
178     validators/report rules; "your video is good enough" UX.
179     6. Multi-sport maturation + recording best practices.
180     7. SaaS / multi-user / async foundations (C12) ? lower until monetization is firm.
181     8. Polish & distribution (C9): installer story, first-run path, codec/output,
economics.
182
183     Recommendation: with the mid-week labeled-video batch, spend ~80% on #1 + #3
184     (flywheel + quality iteration on real data), then unification (#2) and deeper value
185     (#4).
186     ---
187
188     7. Open questions needing owner / counsel sign-off ??
189
190     1. Ultralytics Enterprise price ? only relevant if a permissive detector ever proves
191     insufficient (fallback). Quote-only.
```

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192 2. **Per-checkpoint / dataset licenses** for WASB / TrackNetV3 / any VLM ? train-own or
193 confirm each (C3).
194 3. **Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds** ? confirm live
195 figures with counsel before relying on the "small SaaS under threshold" reading
196 (C5/C6).
197 4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
198 throughput (C7).
199 5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec
200 switch.)
201 ---
202 ## Changelog
203
204 - **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
205 ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
206 "beachhead validation" workstream. Wired the summary into §6.
207 - **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
208 audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
209 removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
210 the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
211 `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
212 exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.
213
214 ---
215
216 ## See also
217 - `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors,
218 beachhead; cited).
219 - `archives/MONETIZATION_AUDIT.md` ? licensing/economics research-of-record (sources +
220 verification transcript).
221 - `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
222 - `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor
223 process).
224 - `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
225 - `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
226 - `GOLDEN_SET_VENDORING.md` ? vendor process + Roora pilot (C8).
227 - `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.

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7. user (2026-06-19T19:17:42.687Z)

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1 # Monetization Dependency Audit ? Hosted Highlights API
2
3 > **Research-of-record (2026-05-30).** This is the original deep-research audit
4 > (per-claim verification transcript + primary sources). For the **current**
5 > commercial status ? what shipped, what's still open ? see the living tracker
6 > [../COMMERCIALIZATION.md](../COMMERCIALIZATION.md), which carries these findings
7 > forward. (Notably: the AGPL/Ultralytics blocker in §1 is now resolved via ADR-005.)
8
9 **Status:** Research complete . **Date:** 2026-05-30
10 **Goal:** Run this project as a **closed-source hosted API** on a cloud **NVIDIA GPU
11 VM** ? users upload badminton videos, receive auto-generated highlight reels ? without legal or
12 economic blockers.
13
14 **Key legal premise (verified):** We do **not** distribute code or binaries to users;
15 everything runs server-side. Therefore **distribution-triggered** copyleft (GPL/LGPL) is **not**
16 triggered. BUT **AGPL §13** (network/SaaS clause) and **external metered-API terms** still
17 apply, and **codec patent royalties** are a separate axis from software licenses.
18
19 **Method:** Deep-research harness ? 5 search angles ? 15 sources ? 48 extracted claims ?
20 adversarially verified (3 independent skeptic votes each; ?2 refutes kills a claim) ? 24
21 confirmed. License findings are **high confidence**, verified against raw `LICENSE` files /
22 official terms pages as of 2026-05. Costs are **order-of-magnitude**. Items needing legal
23 counsel are flagged ??.

```

```

15
16     ---
17
18     ## 1. Risk table
19
20     | Dependency | License | SaaS-hosting triggers obligations? | Verdict | Recommended
action |
21     |---|---|---|---|---|
22     | ultralytics (lib) | AGPL-3.0 | Yes ? §13 network clause; importing the lib
into our app + serving over a network = source-disclosure of the whole app | ? BLOCKER |
Replace (see §2) or buy Enterprise License |
23     | yolov8n.pt weights | AGPL-3.0 (same as code; HF card tagged `agpl-3.0`) |
Yes | ? BLOCKER | Replace weights; swapping codebase but keeping .pt does not escape
|
24     | MonoTrack (shuttle) | Adobe Research License ? noncommercial only | N/A ?
commercial use forbidden outright | ? BLOCKER | Do not use. Use WASB / TrackNetV3 instead |
25     | YOLO-NAS weights | Deci "YOLO-NAS License" ? bans commercial/production/managed-
service use | N/A ? forbidden | ? BLOCKER | Do not use the released weights |
26     | Gemini API (service) | Commercial terms (service, not code) | Paid tier: no
training / no human review. Free tier: yes (trained on + human-reviewed) | ? WATCH |
Must bill the API (paid tier) to keep user video private |
27     | Video codecs H.264 / HEVC / AAC | Patent pools (Via LA, Access Advance, Velos) ?
separate from ffmpeg | Royalties attach to encode/decode, not to ffmpeg's license | ?
WATCH ?? | Emit royalty-free output (AV1/VP9 + Opus); see §5 |
28     | ffmpeg / ffprobe | LGPL-2.1+ (GPL if `--enable-gpl`); not AGPL | No ? no
distribution, no network clause | ? CLEAR | Keep; invoke as subprocess (already do) |
29     | google-genai (SDK) | Apache-2.0 (raw LICENSE + pyproject + PyPI) | No | ?
CLEAR | Keep |
30     | torch / torchvision | BSD-3-Clause | No | ? CLEAR | Keep (audit individual
pretrained-weight licenses) |
31     | opencv-python | MIT (pkg) / Apache-2.0 (OpenCV) / bundled ffmpeg is LGPL | No
| ? CLEAR | Keep |
32     | numpy | BSD | No | ? CLEAR | Keep |
33     | lapx | BSD | No | ? CLEAR | Keep |
34     | fastapi / uvicorn / pydantic / jinja2 / python-dotenv | MIT / BSD | No | ? CLEAR |
Keep |
35
36     Bottom line: Only two real blockers ? (1) Ultralytics (AGPL, code and
weights) and (2) two specific model weights (MonoTrack, YOLO-NAS) we must simply avoid.
Everything else is either already clean or a manageable WATCH.
37
38     > ? Correction to ADR-004's open question: WASB's commercial-use license was the
flagged unknown. WASB-SBDT (`nttcom/WASB-SBDT`) is MIT ? CLEAR for commercial use. The
shuttle path is licensing-safe; only MonoTrack is off-limits.
39
40     ---
41
42     ## 2. Recommended fully-permissive target stack
43
44     ### Player / motion-gating detector (replaces `ultralytics` YOLOv8n)
45     Pick one; all are Apache/BSD for both code and weights:
46
47     | Option | License (code + weights) | Notes |
48     |---|---|---|
49     | torchvision `fasterrcnn_resnet50_fpn` / `retinanet` / `fcos` | BSD | Zero
new dependencies (torch already in stack). Person = COCO class 0 ? exactly what the motion
gate needs. Lowest-risk swap. |
50     | YOLOX (Megvii) | Apache-2.0 (code + COCO weights) | Closest drop-in to YOLOv8
ergonomics; nano?x tiers. ?? repo stale since 2023-02 (maintenance risk). |
51     | RT-DETRv2 (`lyuwenyu`/Baidu) | Apache-2.0 | Real-time, NMS-free transformer;
competitive with YOLO. Use the Baidu repo, not the ultralytics RT-DETR port (that inherits
AGPL). |
52     | D-FINE | Apache-2.0 core + weights | ICLR'25 SOTA real-time. ?? repo bundles
an AGPL YOLO subcomponent ? use only the Apache RT-DETR-derived core. |
53

```

54 ****Recommendation:**** start with ****torchvision Faster R-CNN/RetinaNet**** (no new deps, BSD, trivially clean), move to ****YOLOX/RT-DETRV2**** only if you need more speed/accuracy.

55

56 **### Shuttle tracking**

57 - ****WASB-SBDT**** (`nttcom`, ****MIT****) or ****TrackNetV3**** (`qaz812345` or `alenzex`, both ****MIT****) ? all CLEAR for commercial code use.

58 - ?? Their *distributed pretrained checkpoints / training datasets* are not covered by the MIT code grant ? either confirm per-checkpoint terms or ****train our own**** weights (our roadmap already plans this).

59 - ? ****MonoTrack**** is noncommercial ? exclude entirely.

60

61 **### LLM / semantic rally-boundary step (three viable strategies)**

62 | Strategy | What | License | Trade-off |

63 |---|---|---|---|

64 | ****A) Eliminate**** | Offline temporal action localization ? ****ActionFormer**** (MIT), peers MS-TCN / ASFormer (verify each) | ****MIT**** | Best long-term margin (no per-call cost, no external dependency); matches the Phase-4 offline-segmenter roadmap. Most engineering. |

65 | ****B) Self-host VLM**** | ****Qwen2.5-VL-7B or -32B**** (Apache-2.0) or ****InternVL2.5-8B**** (MIT); native video understanding | ****Apache-2.0 / MIT**** | No per-call fees; pays in GPU VRAM (?16?24 GB for 7B fp16). ?? Qwen ****3B** is `qwen-research` = noncommercial*; 72B is custom-license (commercial OK <100M MAU). |

66 | ****C) Keep Gemini**** | Gemini 2.5 Flash, ****paid tier**** | Service terms | Fastest; ~cheap per video; ongoing metered cost + external dependency. Paid tier required for data privacy. |

67

68 ---

69

70 **## 3. Paid-license fallbacks (only if a swap proves too costly)**

71 - ****Ultralytics Enterprise License**** ? removes AGPL for closed-source/SaaS use of code *and* models. Pricing is ****quote-only / not public**** (confirmed via Ultralytics GH issues #15877, #1888, #9853 ? all routed to sales) ??. Treat as a material recurring cost; only sensible if YOLOv8 accuracy proves irreplaceable.

72 - ****Codec patent licenses**** ? Via LA (AVC), Access Advance / Via LA / Velos (HEVC). Only relevant if we choose to *emit* H.264/HEVC instead of royalty-free codecs (see §5).

73

74 ---

75

76 **## 4. Per-video cost sketch (GPU VM, order-of-magnitude)**

77 Reference workload: ~30-min 1080p match; processing happens on a cloud NVIDIA GPU VM.

78

79 - ****C) Gemini 2.5 Flash (paid):**** input ****\$0.30 / 1M tokens**** (text/image/video), output ****\$2.50 / 1M****. Video tokenizes at ****~263 tokens/sec**** at default resolution ? a 10-min clip ? 158k input tokens ? ****~\$0.05 input**** (+ small output). Per-video API cost is ****cheap****; the economic risk is **volume × latency**, not unit price. Flash-Lite (\$0.10/\$0.40 per 1M) is a cheaper tier; Pro is ~10× pricier.

80 - ****B) Self-hosted Qwen2.5-VL-7B:**** ****\$0 API****, cost = GPU-hours amortized over throughput. Needs 24 GB VRAM (L4/A10-class). Real per-video cost/latency vs. Gemini is an ****open question**** ? measure before committing.

81 - ****A) Offline ActionFormer:**** cheapest at inference (small model, no LLM, no API); cost is the ****one-time training**** + light GPU inference. Best steady-state margin.

82

83 > The exact A/B/C \$/video comparison depends on chosen GPU instance and batch throughput ? ****benchmark on the target VM**** before pricing the product.

84

85 ---

86

87 **## 5. Codec / patent considerations + royalty-free output**

88

89 ****The trap:**** ffmpeg being free (LGPL) does ****not**** grant you the **patent** rights to the codecs it implements. ffmpeg's own legal page explicitly disclaims patent licenses. Patent royalties attach to the ****act of encoding/decoding****, independent of the software license.

90

91 - ****Decoding**** users' uploads (GoPro H.264/****HEVC****) is the unavoidable side. HEVC licensing is ****fragmented**** (Access Advance + Via LA + Velos + independents, ~\$2.80/device combined ? "28× AVC") and is the ****highest-uncertainty**** item ??.

92 - ****AVC/H.264:**** Via LA pool ? first ****100,000 units/yr royalty-free****, ~\$0.20/unit after, annual cap (~\$9.75M), and a permanent *****free to end users over the internet**** ? no royalty** provision. Royalties target encoder/decoder ***products***, not transcoding of content. A small SaaS realistically falls under these thresholds ? but confirm with counsel.

93 - ****Mitigation ? emit royalty-free output:**** generate highlights in ****AV1**** or ****VP9**** video + ****Opus**** audio (e.g. WebM/MP4). This sidesteps all ***encode-side*** AVC/HEVC exposure.

94 - ****SVT-AV1**** encoder = ****BSD-3-Clause-Clear + AOMedia Patent License 1.0**** (royalty-free patent grant) ? clean for commercial server use; ****CPU encoder**** (canonical repo on GitLab).

95 - ****VP9**** (libvpx, BSD) and ****Opus**** (BSD, IETF royalty-free) round out a fully royalty-free container.

96 - ****GPU acceleration:**** NVENC ****AV1** encode requires Ada-class** GPUs (RTX 40, ****L4 / L40****). Ampere (A10/A100) and Turing (T4) can ***decode*** AV1 but ****cannot HW-encode**** it ? use ****SVT-AV1 (CPU)**** on those, while H.264/HEVC encode stays HW-accelerated everywhere.

97

98 ****Recommendation:**** provision an ****Ada-class GPU (L4/L40)**** so AV1 output is hardware-accelerated, and standardize highlight output on ****AV1 + Opus****. This removes the encode-side codec-royalty question almost entirely; only inbound HEVC ***decode*** remains as a low/uncertain residual risk.

99

100 ---

101

102 **## 6. Hosted-API alternatives (for strategy C comparison)**

103 - ****Anthropic (Claude):**** commercial terms ? *****no training on customer Inputs/Outputs****, customer ****owns Outputs****, may power your product; bars competing-model use. ? ****No native video**** (frames must be sent as images) ? weaker than Gemini for this use case.

104 - ****OpenAI:**** no training by default; up to 30-day abuse-retention then deletion; Zero-Data-Retention available.

105 - ****Gemini**** remains the strongest ***video-native*** hosted option; keep it on the ****paid tier****.

106

107 ---

108

109 **## 7. Open questions / needs verification before committing ??**

110 1. ****Ultralytics Enterprise price**** ? get a quote to compare against swap engineering cost.

111 2. ****Pretrained-weights / dataset licenses**** for YOLOX, ActionFormer, TrackNetV3, WASB ? repos state a ***code*** license but don't separately license distributed checkpoints. Train-own, or confirm each checkpoint.

112 3. ****Exact current Via LA (AVC) / Access Advance (HEVC) rates & free thresholds**** ? Via LA's page blocked automated fetch; confirm live figures with counsel before relying on the "small SaaS is under threshold" reading.

113 4. ****Real per-video GPU cost & latency**** of Qwen2.5-VL-7B (L4/A10) vs. Gemini paid tier at expected throughput.

114

115 ---

116

117 **## 8. Primary sources**

118 - Ultralytics License (AGPL-3.0 + Enterprise): ``ultralytics.com/license``, ``docs.ultralytics.com/help/license/``, repo ``LICENSE`` (GitHub API ``spx_id=AGPL-3.0``), weights tag on HF ``Ultralytics/YOLOv8`` (``agpl-3.0``), GH issues #15877/#1888/#9853.

119 - AGPL-3.0 §13 text: ``gnu.org/licenses/agpl-3.0``.

120 - google-genai SDK: ``github.com/googleapis/python-genai`` ``LICENSE`` (Apache-2.0), PyPI metadata.

121 - Gemini terms/data/pricing: ``ai.google.dev/gemini-api/terms``, ``.../docs/data-usage``, ``.../docs/pricing``.

122 - ffmpeg legal: ``ffmpeg.org/legal.html``. Codec pools: Via LA (``via-la.com``), Wikipedia H.264 / HEVC patent-licensing. NVIDIA NVENC matrix: ``developer.nvidia.com/video-encode-and-decode-gpu-support-matrix``. SVT-AV1: AOMedia repo (GitLab).

123 - Detectors: ``Megvii-BaseDetection/YOLOX`` (Apache), ``lyuwenyu/RT-DETR`` (Apache), ``Peterande/D-FINE`` (Apache + AGPL YOLO caveat), ``Deci-AI/super-gradients`` ``LICENSE.YOLONAS.md`` (noncommercial weights).

124 - Shuttle: ``nttcom/WASB-SBDT`` (MIT), ``qaz812345/TrackNetV3`` (MIT), ``alenzex/TrackNetV3`` (MIT), ``jhwang7628/monotrack`` (Adobe Research, noncommercial).

125 - LLM/VLM: ``happyharrycn/actionformer_release`` (MIT); Qwen2.5-VL model cards (7B/32B

Apache-2.0, 3B qwen-research, 72B qwen license); `OpenGVLab/InternVL2\_5-8B` (MIT); Anthropic Commercial Terms; OpenAI Enterprise Privacy.

126

127 *Full verification transcript (per-claim 3-vote verdicts): workflow run
`wf\_ddd29371-8c2`.*

128

8. assistant (2026-06-19T19:17:46.931Z)

Now let me read the PMF market research and the storage-related docs that are clearly relevant to BYO-storage.

9. user (2026-06-19T19:17:47.812Z)

```
1      # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3      > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4      > validate live before relying on them for investment/pricing decisions. This is the
5      > market-research-of-record behind the PMF section of the living
6      > [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker ($6). Product framing:
7      > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8      > self-recorded, **static-camera, full-court amateur** racket-sports video and
9      > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10     > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12     > **Confidence convention.** Figures from named industry reports, federations, or
13     > company pricing pages are tagged with source + year. Inferred items are tagged
14     > **(estimate)** / **(low confidence)**. Emerging-category market sizes vary widely
15     > between vendors; ranges are shown rather than false precision.
16
17     ---
18
19     ## 1. Executive summary
20
21     The product sits at the intersection of three favorable trends: (a) a fast-growing
22     **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
23     depending on vendor); (b) a structural shift to **amateur self-recording** on action
24     cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
25     segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
26     106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
27     2024, +45.8% YoY).
28
29     The central tension is the **static-camera constraint**: the tool works best on fixed,
30     full-court amateur footage and *fails* on broadcast/panning footage. This is both a
31     moat (most competitors target team sports / broadcast) and a filter on the addressable
32     user ? the customer must already set a camera on a tripod behind the court. That
33     behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not
34     the median casual player.
35
36     **Recommended beachhead (strongest single):** **badminton coaching academies and
37     serious club/competitive players in India** ? a large, rapidly formalizing academy
38     ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing
39     willingness to pay for coaching, an underserved tooling gap (incumbents are
40     team-sports/broadcast), and a direct-but-immature competitor field. The product is
41     *native badminton-first* ? a genuine edge here. Secondary: **(2) US/global pickleball
42     players & clubs** (capital-rich, fastest-growing, but SwingVision already strong);
43     **(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).
44
45     **Pricing** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)
46     and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A
47     software-only BYO-camera model undercuts the hardware players; realistic anchors
48     **(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.**
49
50     **Top risks:** (1) amateurs won't reliably set up a tripod ? TAM collapses to
51     coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality
```

variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia, China) are the most price-sensitive and hardest to bill at Western SaaS prices.

2. Market sizing & trends

2.1 Sports analytics / sports-video market

- ****2025 size:**** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business Insights); other vendors USD 5.676.0B. ***(2025)***
- ****Growth:**** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%); Future Market Insights ? ~USD 36.2B by 2035 (22.1%). ***(reports 2024?2025)***
- ****Video analytics is the leading segment**** within sports analytics (Grand View / Fortune BI) ? the directly relevant sub-market.
- ****Interpretation (estimate):**** the **amateur/club auto-highlight** slice is a small but fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption

- ****2024 size:**** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). ***(2024?2025)***
- ****Sports is the largest end-use segment**** of action cameras (GMIInsights, 2024).
- ****GoPro ~47% share, DJI ~20%**** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- ****Implication:**** the capture-side behavior the product depends on is already widespread and growing ? **for the segments that record** ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)
Badminton	220M+ play regularly; 160+ countries BWF est. via BadmintonBites (2023)	
Badminton ? China	~80M registered, up to ~100M (largest base) industry compilations (2023)	
Badminton ? India	20M+ recreational by 2023; academy boom + Khelo India industry compilations (2023)	
Badminton ? Indonesia	National sport; ~1M club players industry (est.)	
Badminton ? Denmark	100,000+ registered club members (Europe #1) industry (2023)	
Badminton ? Japan/Korea	Major school-club + workplace sport industry (2023)	
Tennis	106M players (2024)**; +25.6% over 5 yrs (87M in 2021) ITF Global Tennis Report (2021, 2024)	
Tennis ? Asia	~1/3 of players (33M of 87M) ITF (2021)	
Pickleball ? US	19.8M players (2024)**; +45.8% YoY, +311% over 3 yrs; ~6.2M core SFIA / Pickleheads / The Dink (2024)	
Pickleball ? US courts	~70,000 courts, +55% places to play YoY SFIA / Pickleheads (2024)	
Pickleball ? global	IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim **(low confidence)** various (2024?2025)	
Table tennis	30M+ players; 227 member associations ITTF (current)	
Cricket	2.5B+ fans; 500M+ in India (mostly <i>*fans*</i> , not self-recorders) Cricket Rise / Nielsen (2024?2025)	
Pickleball market value	~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR (Market.us/Custom Market Insights, 2024?2025).	
Badminton equipment market	~USD 7.23B global (2023), Asia-Pacific dominant.	
Geographic read	badminton density is overwhelmingly Asian + Denmark; tennis is broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket is huge as <i>*fandom*</i> but its amateur self-recording + static-full-court fit is weak (large field, not a fixed-frame court). For <i>*this product's static-camera fit*</i> , badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap item, not a beachhead.	

109 ---

110

111 ## 3. Customer segment evaluation

112

113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit
| B2C/B2B | Verdict |

114 |---|---|---|---|---|---|

115 | ****Individual amateur/enthusiast**** | Largest raw TAM (100s of millions) | ****Low**** ?
competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | ****Mixed**** ?
most won't set a tripod | B2C | Volume play, weak economics; use as funnel |

116 | ****Coaches & coaching academies**** | India academy boom; thousands across Asia |
****High**** ? already sell coaching; USD 300?1,500/yr plausible | Reachable via
federations/distributors/social | ****Strong**** ? already film fixed-angle | B2B/B2B2C | ****Best
fit**** ? buyer, recorder, payer aligned |

117 | ****Clubs & leagues**** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD
500?3,000/yr | Medium ? committee sales cycle | ****Strong**** | B2B/B2B2C | Strong secondary;
longer cycle |

118 | ****Tournament/event organizers**** | Many small/regional events | Medium ? per-
event/seasonal | Medium ? episodic | ****Strong**** | B2B | Good per-event upsell; lumpy |

119 | ****Schools & universities**** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-
constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | ****National/regional federations**** | Few, high-value | Medium-high; procurement-
heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | ****Broadcasters / streamers**** | High value per deal | High | Low ? incumbents
entrenched | ****Poor** ? panning = failure mode** | B2B | ****Avoid**** ? violates core constraint |

122

123 ****Key dynamics:**** the static-camera constraint is a ****B2B filter**** ? coaches/academies/
124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest
125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters
126 explicitly violate the constraint ? exclude.

127

128 ---

129

130 ## 4. Competitive & pricing benchmarks

131

132 | Product | Category | Model | Price | Notes (year) |

133 |---|---|---|---|---|

134 | ****Veo (Cam 3)**** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G
~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; \$115M raised (2025) |

135 | ****Trace**** | Soccer auto-filming | ****Free camera + sub**** | USD 180/yr (Basic) ? 300
(Pro) | Camera leased free with sub (2025) |

136 | ****Pixellot (Air)**** | Auto camera/streaming | Hardware + sub | Camera ~USD 949 + USD
69?167/mo | Institutional/league; 10,000+ clubs (2025) |

137 | ****Spiideo**** | Auto-filming/analytics | Sub incl. camera | ~USD 3,800/yr (1 team) ?
6,100/yr (club) | Institutional budgets (2025) |

138 | ****Hudl**** | Team video analysis | SaaS tiers | From ~USD 400/yr | Broad team-sports
incumbent (2026) |

139 | ****SwingVision**** | ****Tennis/pickleball AI, single iPhone**** | Software freemium | ****USD
179.99/yr****; free 2 hrs/mo | Closest analog; 200k+ monthly users; \$10M+ raised (2025) |

140 | ****BadPro+**** | ****Badminton AI on phone**** | ****Pay-per-use, no sub**** | Per-analysis (? "<
one coaching session") | Direct badminton competitor; shot detection, heatmaps, auto-highlights
(2025) |

141 | Niche pickleball/badminton apps | Various | Mixed | Varies | Fragmented; few true
auto-highlight tools (2025) |

142

143 ****What buyers actually pay (synthesis):**** consumer software-only anchors at ~USD
144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,
145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ?
146 signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity +
147 warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and
148 sells worldwide via SaaS. Realistic anchors ****(estimate):** B2C USD 60?180/yr; B2B
149 academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline
150 self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B****
151 value story more than a B2C one.

152

```

153 ---
154
155 ## 5. Recommended beachhead
156
157 Ranked by **WTP × participation density × existing recording behavior × static-camera
158 fit × competitive whitespace.**
159
160 ### #1 (strongest) ? Badminton coaching academies & serious players, **India**
161 20M+ recreational players (2023); rapidly **formalizing academy ecosystem** + Khelo
162 India tailwind; academies already record fixed-angle; coaches are buyers who monetize
163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton
164 AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*.
165 **Watch-outs:** price sensitivity (need INR-tuned tiers); sell to the **academy**, not
166 the individual.
167
168 ### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;
170 strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:**
171 **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+
172 monthly users) ? head-to-head differentiation, not greenfield.
173
174 ### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,
175 Korea) + Denmark**
176 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea
177 school-club & workplace systems), mature club infra, fixed-court recording.
178 **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia
179 price-sensitive; Denmark/EU small but high-WTP and easy to pilot.
180
181 **Bottom line:** lead with **India badminton academies** (whitespace + behavior + buyer
182 alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C
183 motion**, then expand to **pickleball-US** (out-build SwingVision) and
184 **badminton-Asia/Denmark** (density).
185 ---
186
187 ## 6. Risks & open questions (what would invalidate the thesis)
188
189 1. **Tripod/static-camera adoption among amateurs.** If serious amateurs won't set up a
190 fixed camera, B2C TAM collapses to coaches/clubs. *Validate:* observe 3?5 academies +
191 50?100 enthusiasts on whether they already record fixed-angle.
192 2. **Free / near-free alternatives** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193 B2C WTP. *Validate:* fake-door pricing at USD 60?180/yr.
194 3. **Capture-quality variance = the failure mode** (gym lighting, off-center angle,
195 occlusion, mixed resolutions). *Validate:* benchmark rally-detection P/R on 100+ real
196 amateur clips; define a "good-capture" spec + auto-reject (ties to
197 `VIDEO_RECORDING_GUIDELINES.md` + the observability reports).
198 4. **Monetization mismatch in best-participation markets** (India/Indonesia/China are
199 price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy
200 INR licensing.
201 5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports).
202 *Validate:* ship a badminton-quality advantage incumbents can't trivially match
203 (offline self-improving detection, per-rally tagging).
204 6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/
205 academy self-serve before institutional deals.
206 7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real
207 purchasing criterion via 5?10 buyer interviews.
208
209 **Cheapest next steps:** (a) 10?15 discovery interviews with Indian badminton academy
210 owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door
211 pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).
212 ---
213
214
215 ## 7. Sources
216

```

217 - Sports analytics market: Precedence Research; Fortune Business Insights; Grand View
 218 Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
 219 - <https://www.precedenceresearch.com/sports-analytics-market>
 220 - <https://www.fortunebusinessinsights.com/sports-analytics-market-102217>
 221 - <https://www.grandviewresearch.com/industry-analysis/sports-analytics-market>
 222 - <https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp>
 223 - <https://www.futuremarketinsights.com/reports/sports-analytics-market>
 224 - Action camera market: [https://www.gminsights.com/industry-analysis/action-camera-](https://www.gminsights.com/industry-analysis/action-camera-market)
 market (2024);
 225 <https://www.precedenceresearch.com/action-camera-market> (2024);
 226 <https://electroiq.com/stats/action-camera-statistics/> (2024)
 227 - Badminton participation/market: [https://badmintonbites.com/how-many-badminton-players-](https://badmintonbites.com/how-many-badminton-players-are-there-in-the-world/)
 are-there-in-the-world/ ;
 228 [https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-](https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-popular/)
 popular/ ;
 229 <https://en.wikipedia.org/wiki/Badminton> ;
 230 <https://www.marketgrowthreports.com/market-reports/badminton-market-110049> ;
 231 <https://www.thesportsschool.com/top-10-badminton-academies-in-india/>
 232 - Tennis: [https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-](https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-participation-hits-106-million-in-five-years/)
 participation-hits-106-million-in-five-years/ ;
 233 [https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-](https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/)
 gains/
 234 - Pickleball: [https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-](https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/)
 in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-
 court-and-facility-demand/ ;
 235 [https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-](https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/)
 nearly-20-million-in-2024/ ;
 236 <https://www.pickleheads.com/guides/pickleball-statistics> ;
 237 <https://www.custommarketinsights.com/report/pickleball-market/> ;
<https://market.us/report/pickleball-market/>
 238 - Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
 239 - Cricket: <https://cricketrise.com/cricket-fans-in-india/> ;
 240 <https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/>
 241 - Competitors & pricing: <https://www.veo.com/en-us/pricing> ; <https://traceup.com/pricing>
 ;
 242 <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
<https://swing.vision/subscribe> ;
 243 <https://www.badproplusapp.com/>
 244

10. user (2026-06-19T19:17:48.220Z)

```

1      # Storage-priced sharing ? mental model (DRAFT v2 for co-design)
2
3      **Status:** DRAFT ? owner + Claude co-design session pending; nothing here is ratified.
4      **Date:** 2026-06-12 (v2 adds: model-tier axis, social layer, on-device end-state, abuse
5      economics ? per owner's second brief) . **Source:** owner briefs (voice note `Umiya
6      Velociti.m4a`
7      pending transcription) + UI_PROPOSAL . HOSTING_PLAN . PMF research . NEXT_STEPS quality
8      table.
9      ? If ratified, this **supersedes the monetization framing in COMMERCIALIZATION.md**
10     (away from
11     per-video/subscription, toward storage-anchored tiers that bundle model quality) ?
12     record as an ADR.
13
14     ## 1. The one-line model (v2)
15
16     Don't sell videos or subscriptions ? **sell shelf space, and let model quality ride the
17     shelf**;
```

13 free = **2 GB + the owned model**;

14 paid = **more shelf + the best model**;

15 the visible meter is

16 always storage. The social layer makes the shelf worth filling; the owned model makes

17 the free

18 tier cost ~nothing; the end-state moves compute onto the customer's own hardware

19 entirely.

```

16
17     ## 1a. The three coupled axes (one user-visible meter)
18
19     | Axis | Free | Paid | End-state (serious users) |
20     |---|---|---|---|
21     | **Storage** (the meter) | 2 GB library | 10 GB+ / pooled | Mostly local disk; cloud =
shareable library only |
22     | **Model** (rides the tier) | Owned model only ? **never Gemini** | Best path (two-
stage refine / frontier) | Best owned model, on-device |
23     | **Compute** (the cost) | Our GPU, capped runs | Our GPU, priority | **Their GPU**
(USB-drive workflow) |
24
25     ## 2. The load-bearing decision: two tiers of object
26
27     | Tier | What | Size reality | Stored? | Quota? |
28     |---|---|---|---|---|
29     | **Source** (raw match upload) | Processing INPUT | 1?4 GB each | Temporarily: kept
through processing + correction window, then **auto-deleted** (e.g. 7 days or on "corrections
confirmed") | Not by default; optional "keep my raw match" toggle = counts |
30     | **Library** (compiled reels, kept rally clips, thumbnails) | The PRODUCT | reel ?
50?80 MB (3?5 min 720p) | Durable | **Yes ? this is what 2 GB means** |
31
32     Why this split: 2 GB cannot hold even one raw match, but holds **~25?40 reels** ? the
quota is
33     only meaningful against outputs. It also kills the "free CDN" abuse vector for a
storage-priced
34     product: **the library only ever contains pipeline outputs** ? there is no path to park
arbitrary
35     files in it. (Sources additionally pass the existing relevance validator.)
36
37     So: "Do we store the uploaded video?" ? **Yes, transiently ? it's a processing input
with a
38     retention deadline, not a possession.** The reel is the possession.
39
40     ## 3. Eviction policies = a pluggable registry (the repo's own pattern)
41
42     `EvictionPolicy` ABC + registry, exactly like `segmenter_registry`:
43     `fifo` (oldest first) . `largest_first` . `smallest_first` . `least_visited` (view-
counter) .
44     `hard_stop` (no deletion ? block new saves until user frees space or upgrades).
45
46     - **Proposed default: `hard_stop`** ? never silently delete someone's match memories;
FIFO etc.
47     are opt-in conveniences. (Co-design question #2.)
48     - **Grace flow:** over quota ? banner + email with the exact list the chosen policy
would evict ?
49     N-day grace (propose 7) ? evict, logged in an `eviction_log`. Offer download-before-
evict.
50     - `least_visited` needs a decision: owner views, public views, or both?
51
52     ## 4. CUJ after upload (end-to-end)
53
54     ```
55     record ? upload source ? async job (PR-1) ? review/correct (corpus engine, PR-4)
56         ? compile reel(s) ? reel lands in LIBRARY (quota meter ticks)
57         ? share: profile page / unlisted link / private
58         ? source hits retention deadline ? auto-delete
59         ? library overflow ? eviction policy (with grace)
60     ```
61
62     A Gmail-style **quota meter** is permanent UI. Profile = `kheldutra.guru/@handle`: reel
gallery,
63     view counts (feeds `least_visited`), sport tags. Per-reel visibility: public / unlisted
/ private.
64

```

```

65     ## 5. Upload path: "500 MB cap" vs "point at a bucket"
66
67     - **Alpha (tunnel ? local box):** keep the PR-2 chunked endpoint. A 500 MB cap ? 10?15
min of
68     1080p ? fine for pre-trimmed games, **kills full matches (1?4 GB)**. Recommend keeping
the
69     4 GB chunked cap and rate-limiting per user instead; tomorrow's CUJ run will tell us
real file
70     sizes. (Co-design question #3.)
71     - **Beta (hosted):** the bucket IS the upload path ? browser ? object storage
**directly** via
72     signed resumable URLs (PLATFORM P2 `StorageBackend`); the server never proxies bytes;
"max
73     filesize" becomes policy enforced at URL-issue + post-upload validation, not a proxy
limit.
74
75     ## 6. The real bill is egress, not storage (sharing = bandwidth)
76
77     Storing a free user's 2 GB costs **< ?5/month anywhere**. Serving is the risk: on GCS,
egress is
78     ~$0.12/GB ? one mildly viral reel (80 MB x 1,000 views) ? **$10 of bandwidth**; on
Cloudflare R2,
79     egress is **$0**. ? Proposed hybrid: **sources ? GCS** (credits, lifecycle auto-delete
does the
80     retention for free) . **library/serving ? R2** (zero egress; virality becomes free
marketing
81     instead of a bill). Storage-anchored pricing only has margins on zero-egress serving.
82
83     ## 7. Pricing sketch (INR, storage-anchored ? numbers are placeholders)
84
85     Free **2 GB** ? paid tiers, e.g. +10 GB / +50 GB monthly or yearly. Anchors: Google One
is
86     ?130/mo for 100 GB (we are not selling bytes ? we sell *organized, shareable match
memory* ?
87     priced above raw storage but inside the PMF envelope: B2C $60?180/yr). Academy (B2B
overlay):
88     **pooled storage** across students + coach-managed profiles. Processing cost note:
Gemini path is
89     ~$0.05/run ? decide whether the free tier includes unlimited processing or N
matches/month
90     (quota is storage-anchored, but compute isn't free ? soft cap recommended).
91
92     ## 7a. The model-tier axis ? why "free never touches Gemini" is load-bearing
93
94     1. **Free-tier economics:** Gemini ? $0.05/run forever; the owned model is ~$0 marginal
on owned
95     GPU. A free tier on the owned model is *sustainable at scale*; one on Gemini bleeds.
96     2. **The corpus flywheel becomes the business model:** free users correct rallies (PR-4)
?
97     golden corpus grows ? owned model climbs the quality table (heuristic 0.48 ? Gen-0
0.74 ?
98     target ? wrapper's 0.66 on multi-court) ? free tier literally improves every quarter
?
99     more users ? more corrections. The owned-model roadmap
(OWNED_MODEL_IMPLEMENTATION_PLAN) IS
100     the free-tier product engine.
101     3. **Independence & privacy:** free tier immune to provider 503 waves, price hikes, and
terms
102     drift; paid tier uses Gemini only on its paid/private tier per existing guardrails.
103     4. **Honest quality ladder (from NEXT_STEPS):** free = owned (today Gen-0-grade,
improving);
104     mid = two-stage WASB+refine (0.83 clean); top = best available path (wrapper 0.93
clean).
105     Paid stays ahead until the owned model overtakes ? at which point margins improve
again.

```

```

106
107    ## 7b. Tier sketch (names placeholder ? co-design)
108
109    | | **Rally** (free) | **Smash** (paid) | **Court** (academy/pro) |
110    |---|---|---|---|
111    | Library | 2 GB | 10 GB+ | Pooled across students |
112    | Model | Owned only | Best path + priority queue | Best owned, **on-device** |
113    | Processing | N matches/mo cap | Higher cap | Unlimited (their hardware) |
114    | Social | Full | Full | + coach overlay, athlete profiles |
115
116    ## 7c. Social layer ? "Facebook of (match) videos," scoped honestly
117
118    The shelf is worth paying for only if filling it is social. **Interactions are free;
storage is
119    the meter** ? never paywall the social loop itself.
120    - **v1 social (already in the alpha/beta plan):** profile `@handle`, reel gallery, view
counts,
121    share links. **v2 social (post-PMF, big build):** follows, reactions (badminton-
native: ?
122    "clean smash"), comments, tag-a-player (consent-gated), academy groups, "rally of the
week."
123    - Network effect is local-first: the same court's regulars find each other ? academy =
built-in
124    graph seed.
125    - ? Scope discipline: feed/moderation/notifications is a product in itself; it lands
AFTER the
126    alpha proves retention, not alongside it.
127
128    ## 7d. On-device end-state ? the USB-drive vision
129
130    Serious users plug a drive of match videos into **their** Windows/GPU machine; the app
processes
131    locally (this repo already *is* that pipeline) and syncs only reels + metadata to the
cloud
132    library/profile. Cloud cost for the heaviest users ? storage + social only.
133    - **The licensing guardrail becomes the product moat:** Gemini can never ship on-device;
the
134    owned model (Qwen3-VL base, Apache-2.0 / MIT-only deps) CAN. Commercial-clean
licensing was
135    the prerequisite all along.
136    - Staging: academy pilot boxes first (serious, GPU-equipped, exactly the owner's own
setup) ?
137    consumer Windows app later (needs the WSL-free Linux-native/DirectML detector path ?
the
138    detector ABC seam exists for this).
139    - Maps to PLATFORM P3 `local`/`byo_worker` compute backends ? productized.
140
141    ## 7e. Abuse economics (why "serious pay, abusers can't extract value")
142
143    Each axis is self-defending: **storage** ? library accepts only pipeline outputs (no
arbitrary
144    files), sources auto-expire; **compute** ? free tier = owned model (cheap) + N runs/mo
cap +
145    relevance validator rejects non-sport uploads; **bandwidth** ? R2 zero-egress serving +
signed,
146    revocable links (and the egress allowance baked into paid storage pricing per $6:
price/GB =
147    storage + ?·expected-views·reel-size·egress-rate + margin; on R2 the ?-term ? 0 and
becomes
148    margin); **social** ? report/takedown + rate limits. There is no tier where an abuser
extracts
149    more value than they pay for, and no free capability that costs us marginal money at
scale.
150
151    ## 8. Guardrails (carry-overs, non-negotiable)

```

```

152
153 Consent-at-capture for other players (Phase-2 recording rules) · **minors never on
public
154 profiles** and never in the training pool · report/takedown path on public reels · share
links
155 signed + revocable · training-data opt-in stays separate from storage/sharing consent.
156
157 ## 9. Schema deltas (when we build ? not before ratification)
158
159 `users(quota_bytes, eviction_policy)` · `library_objects(id, owner, video_id,
kind?{reel,clip,thumb},
160 bytes, created_at, last_visited_at, visit_count, visibility, storage_ref)` ·
`eviction_log` ·
161 source uploads gain `retention_deadline`. `storage_ref` follows P2's `StorageRef`
(backend id + key).
162
163 ## 10. Co-design agenda (the decisions, in order)
164
165 **Object model & policies**
166 1. Ratify the **two-tier model** (sources transient / library durable)?
167 2. **Default eviction:** `hard_stop` (proposed) vs `fifo`? Grace days = 7?
168 3. Quota counts reels only, or also opt-in kept sources? (proposed: both, sources opt-
in)
169 4. `least_visited`: whose visits count?
170
171 **Tiers & pricing**
172 5. Ratify **"free never touches Gemini"** (owned-model-only free tier)?
173 6. Free tier processing cap: N matches/month = ?
174 7. Tier names/sizes (Rally 2 GB / Smash 10 GB / Court pooled ? placeholders) + INR price
points.
175 8. Egress-baked pricing accepted ($7e formula), with R2-for-library hybrid despite GCP
credits?
176
177 **Upload & serving**
178 9. **Alpha upload cap:** 500 MB vs keep 4 GB chunked? (use the first real CUJ file
sizes)
179
180 **Social & end-state**
181 10. v1 social scope = profiles + share links + view counts ONLY (feed/comments post-
PMF)?
182 11. Minors on public profiles: propose **hard exclusion** in v1 (stricter than the
training-pool
183 rule ? public sharing is a different risk class). Ratify?
184 12. Profile namespace: `@handles` on khelsutra.guru?
185 13. On-device milestone placement: academy pilot boxes first, consumer app after the
WSL-free
186 detector path?
187
188 **Paper trail**
189 14. ADR to supersede COMMERCIALIZATION.md's subscription framing once 1?13 settle.
190

```

11. assistant (2026-06-19T19:17:52.012Z)

STORAGE_SHARING_MODEL.md is directly relevant ? it has the upload-path/egress/bucket discussion. Let me check PLATFORM_ARCHITECTURE.md (the StorageBackend registry) and VIDEO_LOCALITY_MODEL.md, which both bear on where BYO-storage slots architecturally and on egress/compute placement.

12. user (2026-06-19T19:17:52.996Z)

```

1 # Video locality ? where the bytes live (mental model + open questions)
2
3 **Status:** THINKING DOC (owner-requested, 2026-06-12) ? a shared mental model for
4 the upload/storage problem, with the open questions named. **No implementation is
5 greenlit by this doc** (per CLAUDE.md, both tracks stay owner-gated).

```

```

6      **Companions:** `UI_PROPOSAL.md` (B2/B3 upload/download), `HOSTING_PLAN.md`,
7      `PLATFORM_ARCHITECTURE.md` S3 (future StorageBackend/ComputeBackend registries),
8      collector ADR-002 (the frame-identical proxy contract this doc leans on).
9
10     ## 0. The questions that triggered this (and the short answers)
11
12     ***"Do we store the video they upload?"** Yes ? today the chunked-upload flow
13     assembles the full file under `security.upload_dir` (default `output/uploads`) on
14     our box, registers it by MD5, and nothing ever deletes it (retention policy is an
15     acknowledged gap ? BACKLOG). Processing also strews derived artifacts (chunks,
16     frames, trajectories, WSL temp) and, on the Gemini path, uploads ?1280px
17     re-encoded chunks to Google's Files API.
18
19     ***"How would they upload a 10 GB file?"** Today they can't: `/api/upload/init`
20     rejects anything over `security.max_upload_mb` (default **4096 MB**), and even
21     under the cap the file moves as sequential ?90 MB parts through a free-tier edge
22     whose body limit forced that part size. For scale: 10 GB ? 30?45 min of 4K phone
23     video (or ~2?3 h at 1080p); on a typical Indian uplink that's **~1 h at 20 Mbps,
24     ~4.5 h at 5 Mbps** ? before retries. Browser-tab uploads of that size are
25     miserable everywhere; expecting academies to manage S3/GCS buckets is a
26     non-starter. **The 10 GB problem is real and the current design doesn't solve it.**
27
28     ***"Can we make it so there is NO upload?"** Mostly ? because of $1.
29
30     ## 1. The central insight: timestamps are the product; the video is dead weight
31
32     Walk the pipeline and ask what each stage actually consumes:
33
34     - The **Gemini segmenter** never sees the original ? it re-encodes chunks down to
35       **1280 px** (`indexing.ai_chunk_scale_width`) before uploading them to Google.
36     - The **detectors/featurizer** consume downsampled frames and trajectories.
37     - The only consumer of the original full-res bytes is the **highlight compiler**,
38       and only to cut clips ? a pure-ffmpeg operation that can run **wherever the
39       original lives**.
40     - What the customer receives is `{rally intervals, report, reel}`. Intervals +
41       report are **kilobytes**. Given the intervals, the reel can be cut on the
42       machine that already has the original.
43
44     So: **any design that moves the 10 GB is moving the wrong thing.** The question
45     is never "how do we upload the video" but "what is the smallest artifact that
46     must cross the network for this tier of service."
47
48     ## 2. The locality ladder
49
50     From most to least bytes leaving the academy's Windows desktop. Sizes are rough
51     ratios for a 4K source ? measure on a real file before committing to any tier.
52
53     | | What crosses the network | ~Bytes (10 GB source) | What it needs | Status |
54     |---|---|---|---|---|
55     | **L3 ? full upload** (today's flow) | everything | 10 GB (blocked: 4 GB cap) | paid
56     edge tier + resumable uploads + server disk + retention policy | built, capped, wrong for big
57     files |
58     | **L2 ? proxy upload** | a timeline-identical ?1080p re-encode | ~1?2.5 GB | a tiny
59     client prep step (ffmpeg) ? **the exact frame-identical contract we already built and verified
60     as collector `prep-annotation-proxy`** (same fps, same frame count, parity-checked, MD5?MD5
61     manifest) | contract proven; client wrapper + server acceptance = new work |
62     | **L1 ? chunks/candidates upload** | only the ?1280px AI chunks (or just candidate
63     windows) the Gemini path needs anyway | ~0.3?1 GB | client-side chunk+downscale (ffmpeg, CPU);
64     server becomes a **GPU-less orchestrator** | the server half exists (it's the gemini segmenter's
65     own flow, relocated) |
66     | **L1-direct** | chunks go **straight from the client to Gemini** ? zero video bytes
67     through our server | ~0 through us | per-user key custody / metering / cost ceilings (OQ4) |
68     idea only |
69     | **L0 ? fully local** | nothing (calls to Google only if the Gemini path is used) | 0 |
70     the owned on-device model (roadmap north star) or a local GPU + our stack | not alpha-feasible;

```

the long-term answer |

60 | ****L4 ? sneakernet**** | a hard drive | 0 over network | a person (the field associate / owner) | how the pilot actually works today |

61

62 ****The honest "no upload" answer:**** ***strictly*** no bytes leave only at L0 with the

63 owned model (and even L0 leaks ?1280px chunks to Google while the Gemini path is

64 the brain ? say that plainly in consent language, OQ3). The pragmatic near-term

65 answer is ****L2 or L1: don't eliminate the upload, shrink it 5?30x**** ? at which

66 point the existing 90 MB-part flow, the 4 GB cap, and a coffee-break wait are all

67 suddenly fine.

68

69 **## 3. Second-order effects worth noticing**

70

71 - ****L1/L2 make the serving box GPU-less for the Gemini tier**** ? ffmpeg + API

72 orchestration only. Hosting cost collapses; the GPU box stays for WASB/owned-

73 model work. (PLATFORM S3's ComputeBackend split falls out naturally.)

74 - ****Storing the proxy instead of the original**** is ~10x less disk AND a smaller

75 privacy surface; the review UI (alpha screen A4) actually ***prefers*** streaming a

76 1080p proxy to seeking inside a 10 GB original.

77 - ****The correction loop survives**** at every tier ?L1: labels are intervals on a

78 timeline, and the proxy timeline IS the original timeline by construction

79 (the collector ADR-002 invariants: same fps, same frame count, parity-verified).

80 - ****video_id keying:**** the indexer keys by MD5 of the file it processed. At L2

81 that's the proxy's MD5; the proxy-manifest pattern (original_md5 ? proxy_md5)

82 already exists ? but the product needs ONE registered identity rule (OQ2).

83

84 **## 4. The frugality frame ? afford the bills by not burning GPU**

85

86 The runway constraint (owner, 2026-06-12): ****the bills stay personally affordable**

87 **for a long time as long as we don't burn too much GPU.**** The locality ladder maps

88 directly onto that:

89

90 - ****The Gemini serving path costs ~\$0.30?0.60 per hour of video**** in API calls

91 (2026-06-10 quality table) and at L1/L2 needs ****no GPU at all**** ? a CPU box,

92 ffmpeg, and an API key. At pilot scale (a handful of academies, a few sessions

93 a week) that is single-digit dollars a month: affordable indefinitely.

94 - ****GPU spend is the thing to guard.**** WASB / owned-model inference and training

95 stay on the owner's already-paid-for local 2070S; rent nothing until the owned

96 model has a ship-gate-passing reason to be served. L2/L1 turn this from a

97 discipline into a default: ***the serving box has no GPU to burn.***

98 - ****Storage follows the same logic:**** keep proxies, not originals (~10x less

99 disk); settle retention (OQ1) before volume arrives; the free edge tier stays

100 adequate precisely because L2 keeps payloads under the existing caps.

101 - ****Put an explicit cost ceiling per video**** (ties into OQ4): a 7-hour tape or a

102 runaway retry must not surprise-bill. The existing ops learnings (serial

103 uploads, ``ai_max_workers=1``, bounded exponential retry) already point this way.

104

105 **## 5. The academy-visit dividend ? field work builds the product, not just the verdict**

106

107 The C13 visits aren't only a market read; run deliberately, each visit deposits

108 product assets:

109

110 - ****Phase-2 recordings (consented, adults-only) ARE the target corpus.**** The

111 owned model's ship target (0.66-to-beat) lives on exactly the messy multi-court

112 footage class these academies produce ? every show-back visit grows the

113 training set the model is starved for, at zero marginal engineering cost.

114 - ****The show-back is a live dress rehearsal of the product loop:**** footage comes

115 back on a drive (L4), we proxy + process + return a reel ? the same motion

116 L2 will automate. Every friction the associate hits there (file handoff,

117 proxy time, reel delivery) is a pre-paid product-backlog item; write it down.

118 - ****Demo reactions and improvement asks**** (captured verbatim on the answer

119 sheet) are the build backlog, collected by someone already standing in front

120 of the user.

121 - ****Connectivity notes from visits**** answer OQ8 with data instead of guesses.

122

```

123    ## 6. Open questions (the decision backlog this doc exists to hold)
124
125    | # | Question | Why it matters / current lean |
126    |---|---|---|
127    | OQ1 | **Retention policy per artifact class** ? original (only L3 has it), proxy,
chunks, frames/trajectories, Gemini remote files, intervals/labels | intervals/labels are the
asset (keep forever, tiny); originals/proxies need an explicit keep-while-active + delete-on-
request rule; Gemini remote-file GC already in BACKLOG |
128    | OQ2 | **Which file's MD5 is the video's identity** at L2/L1 | lean: the processed
file's MD5 (proxy), with original_md5 carried in metadata ? the collector manifest pattern,
promoted to a product rule |
129    | OQ3 | **Consent language for the Gemini path** ? even "local" modes ship 1280px chunks
to Google | must be plain in the academy-facing terms; paid-tier no-training posture to be cited
|
130    | OQ4 | **Key custody for L1-direct** ? client calls Gemini with whose key? per-user
metering, abuse, cost ceiling per video | unsolved; this is what separates L1 from L1-direct |
131    | OQ5 | **Windows client packaging** ? portable ffmpeg + a signed tiny app vs a
PowerShell script for the pilot | pilot can be a script the associate/owner runs; product needs
an installer story (ffmpeg LGPL build = licensing-clean) |
132    | OQ6 | **What does the review UI play** at each tier | L2 streams the proxy (good); L1
has only chunks ? review degrades ? this quietly argues L2 over L1 as the default |
133    | OQ7 | **Reel delivery** ? server-compiled from proxy (shareable 1080p) vs intervals-
back + client cuts full-res | could be both: proxy reel by default, "premium full-res" cut
locally |
134    | OQ8 | **Real academy uplink numbers** ? the L2-vs-L3 case rests on bandwidth
assumptions | C13 visits can note connectivity informally; don't redesign around guesses |
135    | OQ9 | **Recurring volume** ? an academy recording daily generates 10 GB/day; even L2
accumulates | retention (OQ1) + maybe future delta thinking; not an alpha problem |
136    | OQ10 | **If L3 must survive** (users who refuse any client-side step) | needs
resumable uploads + a paid edge tier + a raised cap; treat as the exception path, not the design
center |
137
138    ## 7. Where this lands (recommendation, not a greenlight)
139
140    **Default the alpha to L2 (proxy upload):** a one-command client prep step built
141    on the *already-verified* proxy contract, uploads that fit comfortably inside
142    today's chunked endpoint and 4 GB cap, a GPU-less serving path (the affordability
143    requirement, $4, satisfied by construction), a review UI that streams the proxy,
144    and intervals that map 1:1 onto the customer's original.
145    **L1/L1-direct is the fast-follow** once OQ4 has an answer; **L0 is the owned-
146    model milestone**, not an alpha decision. The field pilot keeps running on L4 ?
147    a person with a hard drive is a fine storage backend for n=3 academies.
148

```

13. user (2026-06-19T19:17:54.779Z)

```

15:browsable by history**, and choose **where their data lives** (Google Drive / cloud
16:bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
58:| Storage | local `filepath` string; output to `output_dir` | no abstraction; can't BYO
Drive/bucket; raw video must be on the box |
92:    ? Storage (local upload | Google Drive | S3 | ?
109:### 3.1 `StorageBackend` registry
114:class StorageBackend(ABC):
119:    def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
125:    local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`,
126:    Dropbox via `dropboxdrivefs`, reaching for **`smart_open`** as the robust large-MP4
127:    byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as
128:    add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
129:- **Backends:** `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. **BYO-storage**:
130:    the tenant registers their own bucket/Drive; raw video **never leaves their account**.
131:- **Auth ? never persist users' long-lived keys.** For buckets, have the user create an
132:    IAM role you **`AssumeRole`** into, scoped to one bucket/prefix, and mint **presigned
133:    URLs** so raw video moves **browser ? the user's bucket directly, never through our
134:    servers**. For Drive, OAuth with the narrow **`drive.file`** scope; for academies, a
137:    LLM provider can take a **signed URL**. The backend exposes both `download_to_local`

```

```

138: (for compute) and `signed_url` (for the provider/browser), so the pipeline stops caring
169: **storage** stays in the user's bucket (signed URLs) while compute runs on **our** GPUs
235: for its tenant, resolves input via the tenant's `StorageBackend`, runs the pipeline,
303: annotations ? LoRA/QLoRA) regression gate) paid fallback)
401:**Engine (don't reinvent):** ms-swift`** already delivers this for VLMs ? CLI-driven,
412:3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness to
443: **explicitly-contributed, licensed** corpus ? *not* on whatever sits in a tenant's bucket
463: **per-user held-out eval** to prove the adapter helps (don't ship a regression); base +
477:| **P2** | **StorageBackend` registry + BYO-storage** | new ABC+registry;
`local`?`gdrive`/\`s3`/\`gcs`; replace `video_path` with `StorageRef`; encrypted per-tenant creds
|
479:| **P4** | **Rich rally/shot semantic metadata & tagging** *(the next value stage after
solid rally detection + highlights)* | shot type (smash / backhand drive / net), rally
**outcome** (winner / forced / **unforced error**), context (e.g. **open court but error**) ?
query-API filters + `HighlightMetadata` + a **human correction/review loop** feeding the golden
set via the `annotation` registry + `candidate_segments` |
486:that ended in a smash or backhand drive,"" **rallies where the court was open but the
508: tenant's own bucket/Drive and we process via signed URL or their compute, we minimize
518: what merely passes through a tenant's bucket. **Exclude minors' video from the training
564:[Omitted long matching line]

```

14. assistant (2026-06-19T19:17:58.789Z)

PLATFORM_ARCHITECTURE.md §3.1 is the strongest hit ? it explicitly describes BYO-storage. Let me read that section and the surrounding context fully to quote accurately.

15. user (2026-06-19T19:17:59.800Z)

```

1      # Platform Architecture ? Multi-User "Assistant Coach"
2
3      **Status:** ? Strategy / research ? for review. No code yet.
4      **Date:** 2026-06-07 . **Owner:** avidullu
5
6      > ? **Superseded where it disagrees with
[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
7      > (2026-06-09, authoritative).** Specifically: (1) licensing is **MIT/Apache-2.0/BSD**
(not "Apache-2.0 only");
8      > (2) base model is **Qwen3-VL (Apache)** primary / InternVL3 (MIT) fallback ? **Gemma-4
is AMBER/bench-only**, not a
9      > cleared base; (3) the owned VLM is **not** the per-turn "conversational brain" ?
extraction (TAL head ? later VLM)
10     > feeds a per-video **event store + text-to-SQL** query layer ("extract once, query
many"); the VLM-answers-each-turn
11     > design is rejected. The §5A model/base/QA-harness detail below predates that and is
kept for context only.
12
13     **Purpose.** Map the path from today's **single-user, local** tool to a **multi-tenant
14     "assistant coach" platform**: users upload videos, get highlights **stored and
15     browsable by history**, and choose **where their data lives** (Google Drive / cloud
16     bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
17     principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
18     execution/compute, the detection+highlight pipeline, and data-serving ? behind small
19     **pluggable interfaces**, reusing the registry pattern the codebase already uses for
20     segmenters/providers/validators/sports/annotations.
21
22     **Product framing (the north star *and* the v1 line).** This is an **assistant coach**,
23     not just a clipper ? but the strategy is **phased and unambiguous about scope**:
24     - **v1 (concrete, near-term).** **rally detection ? highlight-reel compilation ? per-
user
25     history ? a coach/user correction loop that feeds the golden set** ("assistant-coach
26     *lite"). Essentially today's pipeline made multi-user ? this is the real v1
deliverable.
27     - **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the
28     **owned video-QA model** (§5A; §6 P4?P5).
29     - **North-star vision (strictly future ? *not* v1):** **gait / biomechanics, per-athlete

```

```

30     identifying templates, and deep coaching feedback under a supervising coach** (§6 P6).
31     Inspiring, but explicitly deferred ? and the part that raises the **biometric**
privacy
32     bar (§7).
33
34     The buyer is the **academy/coach** (beachhead in
35     [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two
36     consequences ripple through the design: (1) the data model keeps an **optional coach ?
37     athlete overlay over a generic core** (§4.1) ? not baked in; (2) gait/biometrics is
38     **special-category data**, gated and deferred (§7).
39
40     > **Competitive note (validated by research):** none of Veo / Hudl / SwingVision /
41     > Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
42     > (SwingVision uniquely runs *edge* compute on the user's iPhone). So **"your video,
43     > your storage, your compute" is a genuine differentiator for privacy-sensitive
44     > academies ? and SwingVision proves user-side compute is commercially viable. All four
45     > share the same **teams ? athletes ? clips** data model, which we adopt (§4).
46
47     > Scope: this is a **direction-setting doc**. It does not start implementation; the
48     > first infra slice (an async job model) is the on-hold #16 in
49     > [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md). Commercial constraints
come
50     > from [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
51     > foundations) and `archives/MONETIZATION_AUDIT.md`.
52
53     ## 1. Current state & gaps (what we're evolving from)
54
55     | Concern | Today | Gap for multi-user |
56     |---|---|---|
57     | Identity / tenancy | none | no users, orgs, auth, or isolation |
58     | Storage | local `filepath` string; output to `output_dir` | no abstraction; can't BYO
Drive/bucket; raw video must be on the box |
59     | Data model | `videos` keyed by file **MD5**, `play_segments`, `events`,
`candidate_segments` (SQLite) | no owner/athlete link; **highlights aren't a first-class
persisted entity** (compile writes a file and returns a path); no per-user history |
60     | Execution | **async job model shipped** ? `/api/process` returns **202 + job_id**, a
single-worker `ThreadPoolExecutor` drains the `jobs` table, `GET /api/jobs/{id}` polls status |
single-tenant + local-only; can't dispatch to a user's VM/GPU yet |
61     | Pipeline | **already pluggable** (segmenter/provider/validator/sport/annotation
registries) | reuse as-is ? this layer is the model for the rest |
62     | API | `/api/process`, `/api/query`, `/api/compile`, `/api/config`; no auth | needs
auth, per-tenant scoping, async job + history endpoints |
63
64     Grounding: `backend/infrastructure/database.py` (schema + `PRAGMA user_version`
65     migration ladder, **v5** ? `jobs` table + per-user `user_models` already landed),
66     `backend/main.py` (endpoints + the async job worker ? `/api/process` ? 202 + job_id,
67     `GET /api/jobs/{id}`), the registry pattern in
68     `backend/pipeline/segmenters/base.py::SegmenterRegistry` and
`backend/providers/base.py`.
69
70     ## 2. Target layered architecture
71
72     Decouple into layers with a clean **control-plane vs data-plane** split ? the
73     orchestrator coordinates *metadata and jobs*; the heavy *bytes and GPU work* can live
74     elsewhere (even in the user's own cloud).
75
76     ...
77
78     CONTROL      ? Presentation / API (auth, per-tenant)      ?
79     PLANE         ? Identity & Tenancy (org?coach?athlete?user) ?
80                 ? Orchestration / Jobs (submit?status?result) ?
81                 ?????????????????????????????????????????
82                 ? dispatch                                ? metadata
83                 ?????????????????????????????????????????
84     DATA         ? Execution / Compute ? ? Data / Serving    ?

```

```

85     PLANE      ? (local | BYO-VM | ? ? (per-tenant DB: ?
86                ? hosted GPU)      ? ? videos, segments, ?
87                ? runs the Pipeline: ? ? highlights, jobs, ?
88                ? detect ? highlight ? ? history) ?
89                ?????????????????????????????????????????
90                ? read input / write artifacts
91                ?????????????????????????????????????????
92                ? Storage (local upload | Google Drive | S3 | ?
93                ? GCS | Azure) ? raw video + highlight outputs ?
94                ?????????????????????????????????????????
95    ...
96
97    The two **new pluggable seams** are **Storage** and **Compute/Execution**; everything
98    else either exists (pipeline) or is net-new infra (identity, orchestration, serving). A
99    **model/training layer** ? the owned generative video-QA model ($5A) ? sits atop these,
100    fed by Storage (training corpus), Data (labels), and Compute (training + inference), and
101    is delivered through the existing **AI-provider registry with a paid fallback**.
102
103    ## 3. Pluggable abstractions (extend the existing registry pattern)
104
105    The codebase already proves the model: an `ABC` + a `Registry` singleton with
106    `register/get/list_available` (e.g. `SegmenterRegistry`). Add two more registries with
107    the same shape so they're plug-and-play and testable in isolation.
108
109    ### 3.1 `StorageBackend` registry
110    Replace the bare `video_path: str` that flows through the pipeline with a
111    **`StorageRef`** (a backend id + a key/URI), resolved by a backend.
112
113    ```python
114    class StorageBackend(ABC):
115        name: str
116        def open_read(self, ref: StorageRef) -> BinaryIO: ... # stream large video
117        def download_to_local(self, ref: StorageRef, dst: str): ... # for ffmpeg/cv2/WSL
118        def upload(self, local_path: str, ref: StorageRef): ... # persist
119        def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
120        def stat(self, ref: StorageRef) -> StorageStat: ...
121        def list(self, prefix: StorageRef) -> list[StorageRef]: ...
122    ...
123
124    - **Recommended implementation:** build the backend over **`fsspec`** (one matrix ?
125    local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`,
126    Dropbox via `dropboxdrivefs`), reaching for **`smart_open`** as the robust large-MP4
127    byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as
128    add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
129    - **Backends:** `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. **BYO-storage**:
130    the tenant registers their own bucket/Drive; raw video **never leaves their account**.
131    - **Auth ? never persist users' long-lived keys.** For buckets, have the user create an
132    IAM role you **`AssumeRole`** into, scoped to one bucket/prefix, and mint **presigned
133    URLs** so raw video moves **browser ? the user's bucket directly, never through our
134    servers**. For Drive, OAuth with the narrow **`drive.file`** scope; for academics, a
135    **service account with domain-wide delegation**. Store any per-tenant creds encrypted.
136    - **Why it matters for the pipeline:** ffmpeg/OpenCV/WSL need a **local file**, while
137    the LLM provider can take a **signed URL**. The backend exposes both `download_to_local`
138    (for compute) and `signed_url` (for the provider/browser), so the pipeline stops
139    caring
140    *where* the bytes live.
141    - **Migration path:** introduce `StorageRef` alongside the current local path; a `local`
142    backend wraps today's behavior so nothing breaks; adapters land incrementally.
143
144    ### 3.2 `ComputeBackend` / `ExecutionBackend` registry
145    Where the pipeline actually runs for a given job.
146
147    - **`local`** ? in-process / local subprocess (today's behavior; dev + self-host).

```

147 - `hosted` ? our GPU VM pool (the SaaS default).

148 - `byo_worker` ? the tenant runs a `pull-based worker` (a small agent) on their own VM/GPU. `Critical security property`: the worker `pulls` jobs scoped to its tenant from the control plane, pulls input from the tenant's `own` storage, runs the pipeline, and writes artifacts + results back ? so `the orchestrator` never holds the user's cloud credentials and needs `no inbound access` to their network. This mirrors the `self-hosted CI runner` pattern (GitHub Actions runners use a 50s outbound-only HTTPS long-poll). This pull-based runner is the `cheapest, most credential-safe v1` for BYO-compute.

156 - For users who want the platform to `provision GPUs` in their `own` cloud account on demand (an academy with an AWS account but no standing GPU box), add `SkyPilot` (BYOC: everything launches in the user's accounts/VPCs; creds stay with them). `Modal` is the managed fallback for users wanting zero infra (accepting that raw video then flows through Modal).

161 - `Operational / support burden` (a real adoption risk). For the primary beachhead ? Indian badminton `academies/coaches` ? asking them to `run and maintain a worker` on their own VM/GPU is a meaningful ongoing load (updates, monitoring, firewall/NAT, the "it stopped after our IT change" ticket). So: `default academy users` to `hosted`, and treat `byo_worker/SkyPilot` as a `power-user / enterprise / privacy-sensitive tier`, not the academy default. If/when BYO ships, plan for `auto-updating, self-healing agents + health heartbeats + clear worker status` in the UI, and budget the support cost.

168 - `Middle tier to consider` ? `"zero-infra BYO"`: a managed-but-isolated path where the `storage` stays in the user's bucket (signed URLs) while compute runs on `our GPUs` in a `strictly isolated, per-tenant, ephemeral sandbox` (nothing persists our side). This gives the BYO `privacy` benefit (raw video never persists on our infra) `without` the user running a worker ? a pragmatic default between full-hosted and full-BYO.

174 The pipeline code is unchanged; only `where it's invoked` and `how it gets bytes` vary.

176 `### 3.3 Reuse, don't reinvent`

177 The existing `segmenter` / `provider` / `validator` / `sport` / `annotation` registries are the proven precedent and stay as the `pipeline` layer. The `annotation` registry + `candidate_segments` table are also the natural seam for the `coach-correction / golden -set` loop in the assistant-coach phase (\$6, P4).

16. assistant (2026-06-19T19:18:04.747Z)

I now have a thorough picture. The PLATFORM_ARCHITECTURE.md §3.1 explicitly anticipates BYO-storage (registers their own bucket/Drive). But the specific autosync/watch-folder behavior is the new part. Let me verify the autosync angle isn't already covered anywhere, and check for any "watch" or polling/ingestion-trigger concept across the docs.

17. user (2026-06-19T19:18:08.234Z)

docs\BACKLOG.md:44:- ? `GS-3` ? `HumanVendorProvider` ? concrete adapter for the chosen annotation vendor (submit/poll/fetch + DPA/secure-link controls). `Vendor` selected for the initial pilot: Roora (Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). `Revisit`: once the Roora pilot confirms quality, then build the adapter.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:68:- `Impact`: Breaks when users override output dir; pollutes trees; makes per-video isolation and cleanup harder.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:315:- Follows all conventions: branched from master, one work item, tests first, doc updated, review fixes + response before further polling, will poll for LGTM/merge then sync and next.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:343:`Process rule` (current convention per #157 review): The author (Claude loop) is responsible for code fixes, addressing review feedback, and resolving any merge conflicts by pushing updates to the branch. The reviewer (owner) reviews, provides feedback/LGTM, and performs the merge after LGTM (and CI green). The loop polls for comments/feedback, applies fixes, and continues until the PR is observed as merged=true.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:367:- Branched from master, one work item, tests + doc, will poll for reviews/comments, address feedback by pushing fixes (author loop handles fixes; owner merges after LGTM), continue until observe merged=true.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:374:- Scheduler (task 019ec6932b9e) already stopped per prior owner request; this poll observed merge=true.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:400:- Branched from master (post #160 merge), one work item, tests + doc, will poll for reviews/comments, address feedback by pushing fixes (author loop handles fixes; owner merges after LGTM), continue until observe merged=true.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:407:- Scheduler deleted; poll observed merge=true.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:430:- Branched from master (post #161 merge), one work item, tests + doc, will poll for reviews/comments, address feedback by pushing fixes (author loop handles fixes; owner merges after LGTM), continue until observe merged=true.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:448:- Scheduler (019ec6b06a1f) remains active; loop will poll again for comments/LGTM. No self-merge. Will advance only on observed merge=true, then execute finale per owner instruction in this review.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:460:- Scheduler (task 019ec6b06a1f) deleted after poll observed merge=true.

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:471:- Scheduler (task 019ec6c3eb9e) deleted after poll observed merge=true for #165.

docs\archives\checkpoints\2026-06-scaffolding-complete\README.md:25:When a quality iteration or major milestone completes, a new dated checkpoint directory will be added here (or under `quality-iterations/`) containing the then-current versions of the key living docs + any eval baselines. This creates an auditable trail without polluting the live `docs/` top level.

docs\COMMERCIALIZATION.md:29:Legend: ? done · ? in progress / watch · ? blocker / needs action · ?? needs legal counsel · ? future

docs\COMMERCIALIZATION.md:38:| | *** IN-FLIGHT (in progress / watch)** | | | |

docs\COMMERCIALIZATION.md:67:| **Gemini API** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH | Must bill the paid tier to keep user video private (C4). |

docs\CODE_MAP.md:209:[Omitted long matching line]

docs\CODE_MAP.md:210:[Omitted long matching line]

docs\CODE_MAP.md:216:[Omitted long matching line]

docs\DESKTOP_APP_PLAN.md:72: ? (NEW) per-OS device-insert watcher ? mount path

docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md:95:> polls. See `scratch/PR1\_HANDOFF.md` for the review bundle.

docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md:110:- Update the static UI to poll `GET /api/jobs/{id}` instead of blocking.

docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md:121:- Submit a job ? poll ? DONE with the same segments the sync path produced.

docs\archives\decisions\DECISIONS.md:425: (a) Gen-2 arrives (ms-swift / Qwen3-VL / cloud-GPU training ? an alien dependency tree), OR

docs\EXECUTION_POLICY.md:6:carries, so waiting / polling / aborting / resuming is declared per job, not hard-coded. P1 lives in

docs\EXECUTION_POLICY.md:7:`backend/infrastructure/execution\_policy.py` (`ExecutionPolicy` + `run\_bounded` / `poll\_until`); the two

docs\EXECUTION_POLICY.md:8:unbounded Gemini waits are now bounded in

docs\EXECUTION_POLICY.md:9:`backend/providers/gemini.py` (the file-processing poll under `run\_bounded` with `op\_timeout\_sec`). Remaining slices (P3b?P4)

docs\EXECUTION_POLICY.md:14:1. the file-processing poll ? `while video\_file.state.name == "PROCESSING": time.sleep(10)` (no deadline);

docs\EXECUTION_POLICY.md:24:- ***Jobs can self-schedule.*** ? Yes ? a job that must wait persists its `next\_poll\_at` + progress and

docs\EXECUTION_POLICY.md:42: poll_every_n_sec: float = 10.0 # cadence for polling an in-progress external op

docs\EXECUTION_POLICY.md:61:3. on timeout/in-progress ? **persist progress + `next\_poll\_at` = now + poll_every_n_sec`, set job

docs\EXECUTION_POLICY.md:63:4. the worker loop (and the startup sweep) claims jobs whose `next\_poll\_at <= now` and resumes them;

docs\EXECUTION_POLICY.md:66:This needs **only** two `jobs`-table columns (`next\_poll\_at`, `progress` JSON ? `progress` already exists)

docs\EXECUTION_POLICY.md:69:`waiting` AND `next\_poll\_at <= now`), so a job migrates to whoever is free.

docs\EXECUTION_POLICY.md:83: `op\_timeout\_sec` + retry/backoff; `poll\_until` = bounded readiness poll). `GeminiProvider.analyze\_video`
docs\EXECUTION_POLICY.md:84: now runs the `files.get` processing poll under `poll\_until` and `generate\_content` under `run\_bounded`
docs\EXECUTION_POLICY.md:88:- **P2 ? per-job policy substrate (DB) ? SHIPPED.** `jobs` table `policy` + `next\_poll\_at` columns
docs\EXECUTION_POLICY.md:94: `waiting` + `next\_poll\_at`, worker released) and re-claimed when due. A daemon wake-timer
docs\archives\MONETIZATION_AUDIT.md:26:| **Gemini API** (service) | Commercial terms (service, not code) | Paid tier: **no** training / no human review. Free tier: **yes** (trained on + human-reviewed) | ? **WATCH** | **Must bill the API** (paid tier) to keep user video private |

[Showing results with pagination = limit: 40]

18. assistant (2026-06-19T19:18:12.363Z)

The `DESKTOP\_APP\_PLAN.md` has a "per-OS device-insert watcher" concept (the USB-drive autosync precedent). Let me look at that and confirm there's no cloud-folder-watch concept anywhere. This is the closest "autosync" precedent.

19. user (2026-06-19T19:18:13.432Z)

40 | D6 | **Feasibility is gated on P0.** No app shell, no packaging work starts until the compute spike returns a verdict. | This doc; `decision-rigor-norm` (NUMBERS INFORM, FOUNDATIONS DECIDE) | P0 is the only un-gated deliverable; P1?P3 are owner-greenlit *after* P0 numbers. |
41
42 ## 3. Foundation ? research / prior art `[design]` / `[researched]`
43
44 ### 3.1 The compute reality (the crux) ? what we know vs must measure
45 - **WASB on a discrete laptop GPU is acceptable** `[verified, measured]`: on the owner's **RTX 2070 Super Max-Q (8 GB)**, WASB inference is the pipeline driver at **~3.2?3.4x real-time** (?3.4 GPU-min per footage-min, ~0.055 s/frame), peak VRAM ~5.7?6.0 GB; proxy (NVENC) ~0.2x RT; 4K stitch (CPU libx264) ~16 s/rally; end-to-end ?5x clip length. Source: `current-state.md` `COST\_SCALING.md` checkpoint (6 Boxhill clips, 2026-06-18).
46 - **The typical enthusiast laptop has NO discrete GPU.** A 10-min 4K clip at 3.4x RT is ~34 GPU-min *with* a 2070S. On integrated graphics / CPU-only the wall-clock is unknown and could be 10x+ worse ? possibly **hours per clip**, which would kill the one-click promise.
47 - **Apple Silicon has no CUDA.** WASB/TrackNet weights are PyTorch; PyTorch MPS (Metal) backend *may* run them, but op-coverage and speed on the WASB model are **unverified** `[design]`. CPU fallback on Apple is also unverified.
48 - **Owned-model tie-in.** `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` already states the on-device answer: *train in Python ? export across the boundary* (ONNX / Core ML / ExecuTorch / GGUF) ? a thin device shell runs the artifact with zero Python*, with the standing constraint to **train with export-clean ops**. A smaller, exported, quantized shuttle detector is a credible answer if the CPU/MPS numbers on the current WASB weights are bad. **This makes P0 partly a referendum on whether the desktop product needs the owned-model track.**
49
50 ### 3.2 The de-WSL native runner is already designed-for `[verified]`
51 The detector layer was deliberately built to make this port low-regret.
[`backend/pipeline/detectors/base.py`](../backend/pipeline/detectors/base.py) documents, verbatim, an **"Optional Linux-native path (the main de-risk goal)"**: a `LinuxNativeRunner`/`ONNXRunner` that **"Skip[s] all wsl / _stage / conda activate, load[s] weights locally (torch or onnx)? write[s] the exact same CSV format so `trajectory\_to\_action\_windows` works verbatim."** The WSL plumbing lives in a separate `WslCommandMixin`, explicitly so a native runner **"must not inherit WSL plumbing."** The actual inference math is already isolated in
[`wasb\_infer.py`](../backend/pipeline/detectors/wasb_infer.py) (torch + the WASB-SBDT modules, emitting `Frame,Visibility,X,Y`). **So the port is: lift `wasb\_infer.py` out of WSL, load weights with native torch/onnxruntime per-OS, keep the identical CSV contract.** The windowing brain (`trajectory\_to\_action\_windows`) and stitch are untouched.
52
53 ### 3.3 Compute backends to evaluate in P0
54 | Path | OS | Toolkit | Status |
55 |---|---|---|---|
56 | CUDA-native | Windows-native, Linux+NVIDIA | torch+CUDA (no WSL) | proven in WSL

```

`[verified]`; native = port |
57 | Apple MPS | macOS (Apple Silicon) | torch MPS / Core ML export | **unverified ? P0
must measure** |
58 | CPU | all (the no-GPU laptop) | torch-CPU / **onnxruntime** (often fastest on CPU) |
**unverified ? P0 must measure; the gating number** |
59 | Smaller/exported model | all | ONNX/Core ML/ExecuTorch, quantized | deferred to owned-
model track; P0 informs whether it's *needed* |
60
61 ### 3.4 App-shell prior art & licensing landscape `[researched]`
62 - **App shell options:** **Tauri** (Rust + system webview; ~3?10 MB binaries;
MIT/Apache) vs **Electron** (bundled Chromium; ~100?150 MB; MIT) vs **CLI + installer** (wrap
`generate_highlights.py`; smallest effort). All shell licenses are clean.
63 - **Backend packaging:** the Python/FastAPI backend bundles via **PyInstaller** or a
frozen venv as a local "sidecar." Packaging was already modernized ? **PEP 621 `pyproject.toml`
+ an `indexer` console entry point** shipped (`DEFERRED_HARDENING_PLAN` #7, DONE, PR #167)
`[verified]` ? so a CLI MVP has a real head start.
64 - **ffmpeg licensing (a genuine landmine):** ffmpeg is LGPL/GPL. The current stitch path
uses **libx264 (GPL)** . Redistributing that inside an installer would impose GPL on the bundle.
**Mitigation (D4):** ship an **LGPL ffmpeg build** using **openh264 (BSD)** or the OS hardware
encoder (NVENC / Apple **VideoToolbox** / Linux **VAAPI**) ? the desktop app should prefer
hardware H.264 anyway (faster, no CPU stitch tax).
65 - **Weights:** WASB-SBDT = MIT `[verified]` (`tracknet-wasb-wsl2-decision`),
redistributable.
66
67 ## 4. Design / architecture `[design]`
68
69 ### 4.1 The one-click flow (and what code each step reuses)
70 ```
71 [card inserted]
72 ? (NEW) per-OS device-insert watcher ? mount path
73 ?
74 [list videos on the card] ? glob video exts; reuse _VIDEO_EXTS from
generate_highlights.py
75 ?
76 [make on-device proxy] ? ffmpeg downscale (hardware encoder); reuse the proxy +
--detect-from split (#205)
77 ?
78 [detect: WASB native] ? (NEW) NativeWasbRunner (DetectorRunner subclass) ? NO WSL;
same CSV contract
79 ?
80 [windowing ? rallies] ? REUSE trajectory_to_action_windows + improved milestone
windowing (R1: cap=6 + R4)
81 ?
82 [stitch reel from ORIGINAL] ? REUSE VideoCompiler
(backend/pipeline/stitcher/compiler.py); read original in place
83 ?
84 [export reel + tiny report to a user folder] ? reuse _atomic_publish non-destructive
write; NEVER copy originals
85 ```
86 The detector runs on a **proxy** ; the stitch reads the **original on the card** (the
`--detect-from` split, already shipped `[verified]` in `generate_highlights.py`). Rally
timestamps transfer 1:1 because the proxy shares the timeline (the collector frame-identical
proxy contract, `VIDEO_LOCALITY_MODEL` §1).
87
88 ### 4.2 Reuse map ? what's NEW vs REUSED
89 | Concern | Status | Code |
90 |---|---|---|
91 | Trajectory?rally windowing | **REUSE** `[verified]` |
`tracknet_runner.trajectory_to_action_windows` (model-agnostic staticmethod) |
92 | Stitch / reel compile | **REUSE** `[verified]` |
`backend/pipeline/stitcher/compiler.py::VideoCompiler` (pure ffmpeg) |
93 | Local orchestration (proxy?detect?stitch, non-destructive, resumable, `--skip-ai-
handoff`, `--detect-from`) | **REUSE** `[verified]` | `tools/generate_highlights.py` |
94 | Detector runner contract | **REUSE** `[verified]` |
`backend/pipeline/detectors/base.py::DetectorRunner` ABC |

```

```

95      | WASB inference math | **REUSE (lift out of WSL)** `[verified]` |
`backend/pipeline/detectors/wasb_infer.py` |
96      | **Native (de-WSL) WASB runner, per-OS** | **NEW** `[design]` | `NativeWasbRunner` ?
sibling of `wasb_runner.py`, no `WslCommandMixin` |
97      | **USB/SD insert detection + video listing** | **NEW** `[design]` | per-OS: Win
WMI/`WM_DEVICECHANGE`, macOS DiskArbitration, Linux udev/udisks2 |
98      | **App shell / one-click UI** | **NEW** `[design]` | Tauri vs Electron vs CLI ($3.4; P2
decides) |
99      | **Packaging + code-signing per OS** | **NEW** `[design]` | PyInstaller/MSIX (Win),
.dmg+notarize (mac), AppImage/Flatpak (Linux) ? P3 |

```

20. assistant (2026-06-19T19:18:23.344Z)

I have a complete picture now. The DESKTOP_APP_PLAN.md establishes the "watcher ? auto-prepare" pattern but for **physical SD-card/USB insert** ? the exact local analog of the proposed **cloud-folder watch**. The cloud-watch/autosync variant itself is not anticipated. I have everything I need to answer.

Key findings synthesized:

- **PLATFORM_ARCHITECTURE.md** \$3.1** explicitly anticipates the **storage half** (BYO bucket/Drive/S3/GCS/Azure, signed URLs, never-persist-keys, `drive.file` OAuth, browser?bucket-direct). It does NOT cover the **watch/autosync trigger** ? that's net-new.
- **STORAGE_SHARING_MODEL.md** \$5-6** has the "point at a bucket" upload path and the egress economics (the real bill is egress, not storage; sources?GCS, library?R2).
- **VIDEO_LOCALITY_MODEL.md** is the cost-story backbone (the locality ladder; "any design that moves the 10 GB is moving the wrong thing"; L1-direct = chunks straight client?Gemini, zero bytes through us).
- **DESKTOP_APP_PLAN.md** \$4.1** has the device-insert watcher ? auto-prepare flow ? the local twin of cloud-folder autosync.

21. user (2026-06-19T19:19:18.293Z)

Structured output provided successfully

22. assistant (2026-06-19T19:19:33.879Z)

I've completed the assessment and returned the structured object. Summary of what I found:

Scope read: **PLATFORM_ARCHITECTURE.md** (\$3 pluggable Storage/Compute registries ? the load-bearing doc), **COMMERCIALIZATION.md** (\$6 PMF/dashboard), **PMF_MARKET_RESEARCH.md**, **archives/MONETIZATION_AUDIT.md**, plus the storage/egress/autosync threads those docs point to: **STORAGE_SHARING_MODEL.md**, **VIDEO_LOCALITY_MODEL.md**, **DESKTOP_APP_PLAN.md**.

Verdict ? partial anticipation. The docs already design the **storage half** of the idea but not the **autosync half**:

- **PLATFORM_ARCHITECTURE.md** \$3.1** explicitly has BYO-storage: `local / gdrive / s3 / gcs / azure` over `fsspec`, "the tenant registers their own bucket/Drive; raw video never leaves their account," signed URLs browser?bucket-direct, `drive.file` OAuth, never-persist-keys. It even names BYO-storage as a competitive differentiator (no Veo/Hudl/SwingVision/Spiideo offers it).
- **What's missing:** no watch/auto-sync/ingest-trigger concept exists anywhere (grep confirms ? only DESKTOP_APP_PLAN's **physical device-insert** watcher and unrelated job-polling). The "watch the folder, pull new recordings as they land, auto-process" component is net-new.

Where it slots: a new Watcher/IngestTrigger layer on top of the existing `StorageBackend` registry ? poll `list()`/`stat()` or use provider push (S3 EventBridge / GCS Pub/Sub / Drive changes-watch / Dropbox webhooks), dedupe by the existing `video\_id`=MD5, submit the shipped async `/api/process`?202 job. Compute placement picks among \$3.2 tiers ("zero-infra BYO" = their bucket + our ephemeral GPU sandbox is the natural default).

The honest gaps the docs do not yet cover: who-pays-egress when WE pull vs the desktop pulls (STORAGE_SHARING \$6 only models **our serving** egress, not **us pulling** from their cloud*), OAuth scope creep (a watcher needs broader read than the privacy-minimal `drive.file` + app-verification + refresh-token custody), streaming-vs-download per stage, retention/consent for

the brief window we hold their video (OQ1/OQ3), and HEVC/format variety hitting a no-human-in-loop firehose.

Relevant absolute paths: `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM\_ARCHITECTURE.md`, `...\docs\STORAGE\_SHARING\_MODEL.md`, `...\docs\VIDEO\_LOCALITY\_MODEL.md`, `...\docs\DESKTOP\_APP\_PLAN.md`, `...\docs\COMMERCIALIZATION.md`, `...\docs\PMF\_MARKET\_RESEARCH.md`.